C

CAD/CAM (*see also* Machining)

CAD/CAM is an acronym for computer-automated design and computer-automated machining. CAM is generally conducted in conjunction with computer numerical control (CNC) for metal-working processes such as blade-tip robotic welding, CNC tig welding, and CNC lathing, milling, and/or machining. All these numerical and computational methods contribute to a field that is now a science in its own right. *See* Metallurgy for information on blade-tip robotic welding, an example of a CNC CA (in this case welding) process.

CFD (Computational Fluid Dynamics)*

CAD is frequently used in conjunction with computational fluid dynamics (CFD) where air- or gasflow is involved. For illustrative purposes, some literature on CFD, as a component of CAD, follows.

CFD can be used for analyzing, for instance

- Fluid handling
- Measurement and controls
- Heat and mass transfer
- Filtration
- Distillation
- Mixing
- Separation
- Fluidization
- Sedimentation
- Reaction
- Polymerization
- Drying
- Forming
- Ventilation
- Emission control
- Incineration
- Combustion
- Materials processing

*Source: Fluent Inc., USA. Adapted with permission.

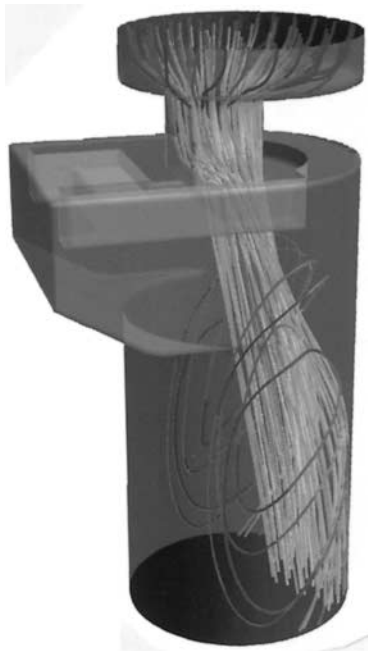


FIG. C-1 Streamlines depict the flow of regenerated catalyst through a slide valve, revealing the source of erosion problems. Courtesy of REMOSA. (Source: Fluent Inc.)



FIG. C-2 Instantaneous solids concentration in a riser, showing how a baffle plate can be used to distribute flow and produce uniform delivery of catalyst to the reactor. (Source: Fluent Inc.)

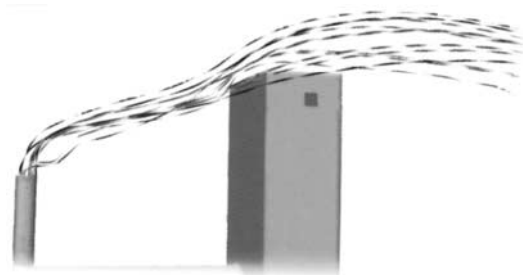


FIG. C-3 Prediction of exhaust plume trajectories can provide quantitative information about downstream impacts. (Source: Fluent Inc.)

Figures C-1 through C-11 are modeling representations of eight individual case studies.

The work done by CFD firms is frequently unique to a specific firm and any research alliances it may have formed with individual OEMs.

Some Generic CFD Applications

Fluid handling and flow distribution

Transport and storage of gases, liquids, or slurries represents a large capital and operating expense in process plants. CFD software helps to design for flow uniformity, balance flows in manifolds, minimize pressure drop, design storage tanks, and accurately size blowers, fans, and pumps. High-speed nozzles and spray systems can be analyzed in order to optimize performance.

Reactor modeling

CFD software helps you to quantify residence times, mixing rates, scaling effects, and overall chemical conversion in a wide range of reactor systems, including packed beds, fluidized beds, recirculating beds, plug flow or tube reactors, and stirred tank reactors. This provides the flexibility you need for description of reactions and the sophistication you need for prediction of gas-solid, gas-liquid, or liquid-solid multiphase systems.

Ventilation and safety

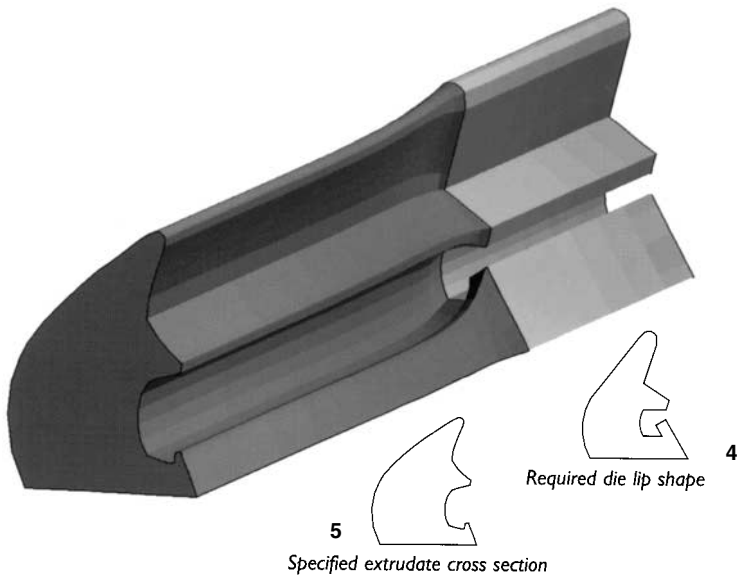
CFD software allows you to reliably and easily determine the trajectory of environmental releases and examine building ventilation system performance. You can quantify the exposure of personnel to specific contaminant levels and analyze the effectiveness of planned responses.

Mixing

Mixing in agitated vessels, static mixers, jet mixers, t-mixers, and other devices is important to the performance of most chemical and process plants. The capability for the analysis of stirred tank mixers is unsurpassed, with interactive automated model generation and mixing-specific data analysis tools.

Materials processing

Extrusion, mold filling, fiber forming, thermoforming, and coating processes involve complex fluid rheology and deforming free surfaces that not all CFD codes can



Prediction of extrudate shape. Inverse die design capability allows determination of the required die lip (FIG. C-4) for a specified extrudate shape (FIG. C-5). (Source: Fluent Inc.)

model. CFD software includes powerful tools that are customized to excel for these applications.

Separation and filtration

CFD can provide a complete range of tools for modeling of phase separation, solids settling, and particle dispersion and classification. Inertial separation using chevrons or cyclonic separators and filtration systems using filter media can also be modeled.

Combustion, incineration, emissions, and environmental control

CFD can provide state-of-the-art models for prediction of combustion and pollutant formation, including built-in NO_x prediction. Optimization of environmental control equipment, from incinerators to scrubbers, filters, and collectors, can help ensure compliance and reduce capital costs.

Some Examples of Specific CFD Case Studies

This material is proprietary to the information source company and therefore contains mention of trademarks specific to this designer.

Process industry modeling

The European Commission has funded the European office of this information source to work on two projects related to process industry applications.

OLMES is a project that looks at the application of CFD to the design of membrane separation devices used in the production of reformulated gasoline. To date, the primary method available for olefin reduction in FCC-derived gasoline is

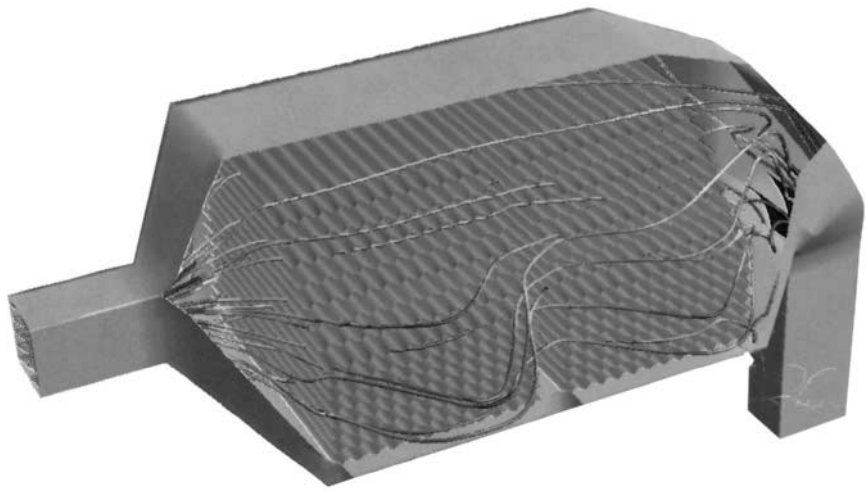


FIG. C-6 Flow streamlines in an electrostatic precipitator confirm the effectiveness of vanes for improving uniformity of flow through the precipitator plates. (Source: Fluent Inc.)

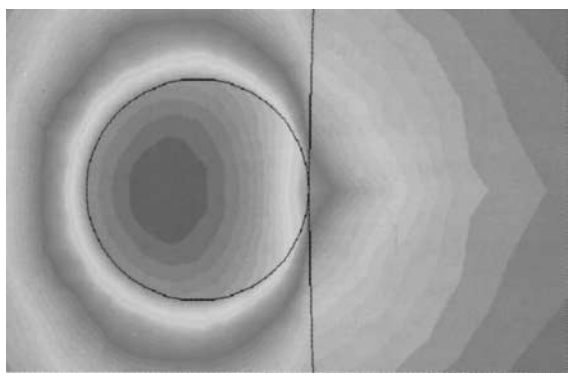


FIG. C-7 Temperature prediction in the vicinity of two catalyst-impregnated particles, studied as part of the CATAPOL project on the application of CFD to polymerization. (Source: Fluent Inc.)

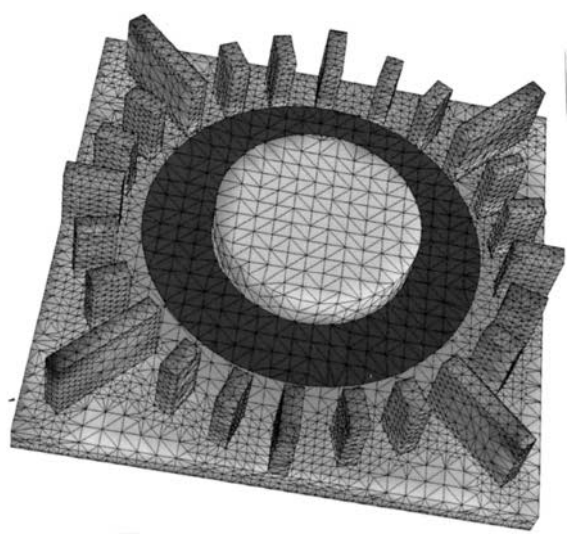


FIG. C-8 Tetrahedral mesh used in IcePak 2.2 to model a complex fan/heat sink with radial fins. (Source: Fluent Inc.)

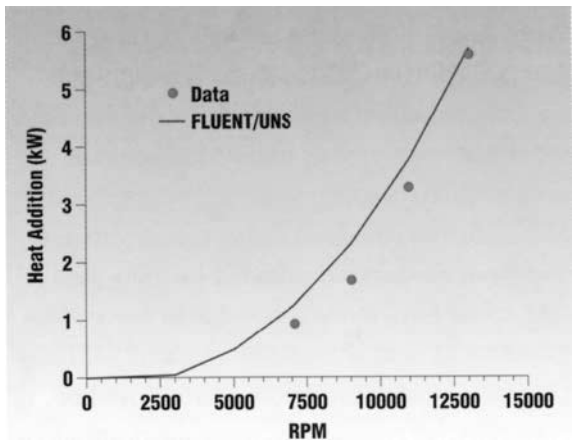
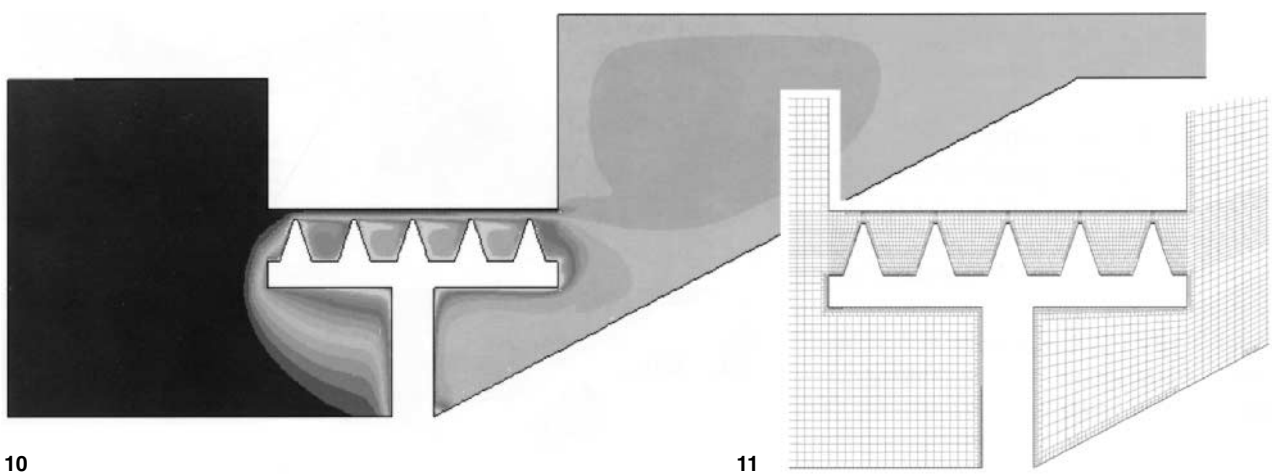


FIG. C-9 The FLUENT/UNS prediction yielded good agreement with the measured heat addition due to windage heating. (Source: Fluent Inc.)



10 11

FIG. C-10 Temperature prediction showing the effect of windage heating. (Source: Fluent Inc.)

FIG. C-11 Mesh adaption with embedding was used to ensure grid independence of the predictions. (Source: Fluent Inc.)

the hydro-treating process. However, hydro-treatment is a very energy-intensive process and significantly reduces octane quality. Membrane separation is now becoming an attractive alternative, but many practical design problems remain. The OLMES project aims to build simulation software that will aid the design process by modeling the fluid flow and mass transfer in the many fine passageways that make up a membrane separator.

CATAPOL is a project that applies CFD to the modeling of heterogeneously catalyzed gas-phase polymerization, involving the injection of catalyst particles into a fluidized bed where they react with monomer gas to grow polymer particles. Using CFD modeling of individual particle behavior and fluidized bed hydrodynamics, the project addresses potential problems such as reactor stability and thermal runaway.

New core technology for IcePak

IcePak, a specialty CFD product for electronics cooling applications, continues to evolve at a fast pace. Jointly developed by this information source and ICEM CFD, new releases of IcePak are delivering on the commitment to make this information source's latest core CFD technology available for electronics thermal management.

The first adaptation of this product, "Version 2.1," released in October 1997, delivered speed improvements by incorporating the solver engine from FLUENT/UNS (an average of 7.5 times faster during regression testing).

The next adaptation of this product, "Version 2.2," was intended to deliver more flexible model building capability, with tetrahedral as well as hexahedral meshes supported. Tetrahedral meshes can handle extremely complex geometries. The combination of automated hex and tet meshing gives IcePak users better strategies when confronted with difficult modeling.

Predicting windage heating in labyrinth seals

Labyrinth seals are commonly used in rotating equipment to restrict cooling flow between rotating and stationary components. One of the issues in the design of such seals is the accurate prediction of the temperature rise of the cooling flow due to windage heating effects. Accurate prediction of this heating allows designers to maintain the structural integrity of the engine with the minimum amount of cooling flow, thereby maximizing the efficiency of the engine.

In order to validate the accuracy of this information source's product ("FLUENT/UNS") for this kind of flow prediction, a straight five-fin labyrinth knife seal with a nominal clearance of 1.11 mm between the labyrinth seal and the shroud was analyzed. The computational model was axisymmetric, with specification of the pressure ratio (1.5) across the seal as imposed boundary conditions. The working fluid was assumed to be air, with density computed via the ideal gas law and fluid properties (viscosity, thermal conductivity, and heat capacity) expressed as a function of temperature. Turbulence was modeled using the RNG k-ε model. The CFD model was run at several different rotor speeds and the windage heating was computed and compared with the experimental data.

Grid independence using adaption. Grid-independent solutions were obtained at each operating speed by using the solution-based mesh adaption capability in FLUENT/UNS. The mesh was adapted based on predicted y^+ values and on gradients of total temperature. The initial mesh for each simulation started out with approximately 9400 quadrilateral cells and after adaption the final cell count was approximately 12,500 cells. A typical mesh after the solution-adaptive refinement is shown below.

Carbon; Carbon-Graphite Mix Products

When mixed with graphite, carbon has lubricant-, strength-, and temperature-resistance properties that are useful in many applications. The properties of the mix depends on the variation in the mix, as well as the bonders and adhesives used. These properties will therefore vary between companies that make these products. For illustrative purposes, one company's product line, with applications, is outlined here.*

*Source: Advance Carbon Products, USA. Adapted with permission.

TABLE C-1 AD-CARB Materials

Grade	Composition	Density, g/cm^3	Shore Hardness	Compressive Strength, psi	Flexual Strength, psi	C.T.E., in/in/°F × 10^{-6}	Modulus of Elasticity, psi × 10^{-6}	Temperature Limit in Air, °F
AC-1	CG	1.59	40	8,000	3,600	2.5	1.7	700
AC-1F	CG-R	1.75	55	11,000	5,000	3.0	1.9	400
AC-2	CG	1.71	60	11,000	6,700	2.5	2.0	700
AC-2F	CG-R	1.80	70	15,000	8,000	3.0	2.2	400
AC-3	CG	1.66	50	12,000	4,600	2.4	1.7	700
AC-4	CG-R	1.75	65	20,000	6,000	2.9	1.9	400
AC-8	G	1.76	40	9,000	5,000	1.6	2.0	750
AC-9	G	1.64	35	4,000	1,900	2.2	1.6	750
AC-10	G	1.72	35	5,600	3,500	2.1	1.8	750
AC-20	G	1.78	45	9,300	4,300	2.3	1.6	750
AC-22	G	1.71	35	7,200	3,500	2.2	1.5	750
AC-23	G-P	1.71	35	7,200	3,500	2.2	1.5	750
AC-25	G	1.72	40	7,000	4,000	1.6	1.6	750
AC-26	G	1.70	45	8,000	4,500	2.6	1.5	750
AC-27	G	1.90	85	18,000	10,000	3.5	1.9	750
AC-30	G	1.81	45	9,000	4,800	1.4	1.5	750
AC-40	CG-B	3.00	40	13,000	6,000	2.3	2.9	400
AC-50	CG-BR	2.85	45	17,000	7,500	2.6	2.9	500
AC-52	CG-BR	2.55	60	25,000	8,500	2.6	3.1	500
AC-60	CG-CU	2.85	50	16,000	7,500	3.1	2.8	750
AC-62	CG-CU	2.50	60	24,000	8,500	3.0	2.8	750
AC-80	CG-AG	3.40	45	16,000	7,000	3.1	2.7	750
AC-82	CG-AG	2.60	65	24,000	8,000	3.0	2.9	750
AC-101	CG-NI	2.40	60	23,000	7,500	2.8	2.8	750
AC-106	CG-NI	2.85	50	15,000	6,500	2.7	3.1	750
AC-236	CG-R*	1.85	85	26,000	9,500	2.9	3.2	500
AC-240	CG	1.70	85	30,000	9,000	2.1	2.5	700
AC-243	G	1.85	90	33,000	12,000	2.8	3.0	850
AC-245	CG-AG	2.45	75	36,000	12,000	2.8	4.0	700
AC-246	CG-R	1.80	95	40,000	12,500	3.2	3.3	500
AC-250	CG	1.75	65	20,000	7,500	2.0	2.3	700
AC-253	CG-AT	2.10	75	27,000	10,000	2.3	3.9	700
AC-266	CG-BR	2.40	80	40,000	13,000	3.7	4.1	500
AC-270	C-B	2.40	65	26,000	10,000	3.7	3.0	400
AC-540	G	1.77	50	13,000	5,500	1.8	1.3	750
AC-562	G-P	1.88	70	21,000	14,000	4.3	1.6	850

Code: CG = Carbon-Graphite; G = Graphite; P = Purified; R = Resin; B = Babbitt; BR = Bronze; CU = Copper; AG = Silver; AT = Antimony; NI = Nickel-Chrome; * = Pressure Tight after Machining

Example of a Carbon Mix Material

AD-CARB is a self-lubricating, low-friction carbon material that has high mechanical strength and can withstand temperatures up to 750°F without oxidizing.

Through the impregnation of metals and/or resins, this material can solve many lubrication problems. See Table C-1.

Product properties

1. Self-lubricating. The product acts as an effective dry lubricant, forming a thin film on mating parts, which allows the material to properly function without additional lubrication. The product may be used with light lubrication to enhance its load-carrying capabilities. The material is nongalling.

2. Oxidization resistant. The product maintains its properties up to 750°F without oxidizing. The material may be used up to 1500°F in an inert atmosphere, depending upon which grade is used. Certain materials may be treated to operate successfully up to 1100°F in an oxidizing atmosphere.

3. High mechanical strength. The product possesses high mechanical strength and hardness to create a long-wearing material. The strength of the material actually increases with an increase in temperature.

4. Chemical resistant. The product is chemically stable and does not react with most chemicals: acids, alkalis, salts, and organic solvents.

5. Low coefficient of friction. The product has a coefficient of friction of approximately 0.05 to 0.20, depending upon the lubrication and surface finish of the mating surface.

6. Low coefficient of thermal expansion. The product has a coefficient of thermal expansion of between 1.5 and 4.0×10^{-6}/°F, allowing for precisely controlled running clearances from room temperature up to maximum operating temperature.

7. Material classification. The product is a mixture of carbon and graphite that may be impregnated with various metals and resins to enhance the wear properties and improve the life of the materials.

Design factors

1. Environment. The carbon material and its impregnation must be checked to ensure that they are compatible with the environment.

2. Temperature. Though carbon does increase in strength with an increase in temperature, the maximum operating temperature of the material must not be exceeded. When the maximum temperature is exceeded, carbon becomes soft on the exposed surfaces from oxidization and will wear quite rapidly.

3. Mating materials. The product may be run against most materials with better results when run against the harder surfaces. Ceramics and chrome-plated surfaces are excellent mating materials. A minimum Rockwell C of 55 is recommended. Longer life will also be obtained if the mating material is ground and polished to a 16 microinch finish or better.

4. Load. The bearing load, in pounds per square inch, is calculated by figuring the total bearing load divided by projected area of the bearing (length × diameter). The maximum load that a bearing can withstand is related to the velocity (feet/minute) by the following formula: PV = 15,000 (dry).

If the bearing is thoroughly lubricated, the PV may be increased up to 150,000 or more, depending upon the circumstances. See Fig. C-12.

5. Interference fit. Since carbon bearings are usually installed in metal housings and run against metal shafts, the designer must be aware of the difference in the coefficient of thermal expansion of the materials and take this into consideration when designing the size of the bushing. At maximum operating temperature, the bushing must have at least a 0.0015 in interference fit per inch of diameter.

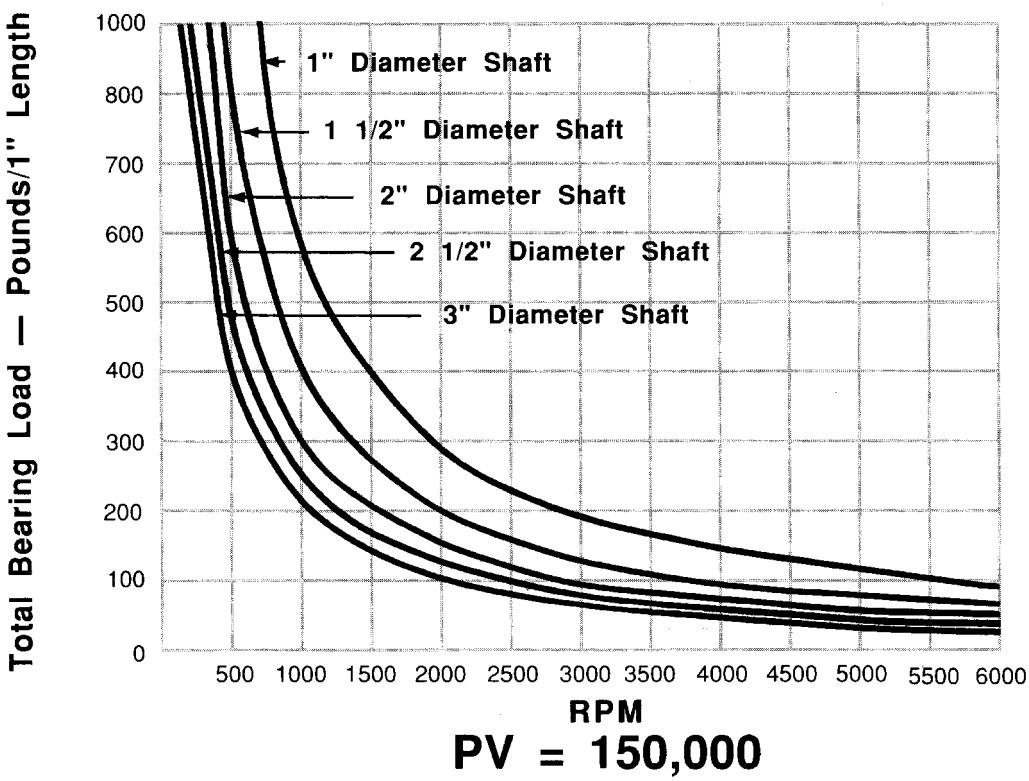


FIG. C-12 Maximum allowable load for thoroughly lubricated bearings. Reduce load to 10 percent of graph value for dry running. (Source: Advance Carbon Products.)

6. Close-in. When a carbon bearing is either pressed into a housing or heat-shrunk into a housing, the inside diameter will shrink in relation to the amount of interference on the outside diameter. This shrinkage must be taken into account when designing the inside diameter. For heavy-wall bearings (approximately ½ in), the inside diameter will shrink approximately 40 percent of the interference fit on the outside diameter; for thin-wall bearings (approximately ⅛ in), the inside diameter will shrink approximately 90 percent of the interference fit on the outside diameter.

7. Running clearance. The running clearing will vary depending upon the application requirements; however, a general guideline is to have 0.002 in/in of shaft diameter with a minimum of 0.003 in. This minimum clearance is to allow foreign particles to escape through the bushing, with minimal damage. These clearances are measured after the bearing is installed in the housing to take into account the inside diameter close-in. With high surface speeds the clearance may be reduced, and with slow surface speeds the clearance may be increased.

8. Bearing length. The bearing length should be one to three times the shaft diameter, with 1½ times the preferred length.

9. Preferred design. The preferred design is for the bearing to be a straight sleeve type without shoulders and grooves. Shoulders create stress points and can be a cause of failure. Grooves can be used quite successfully if they are installed with a radius tool bit and do not cut so deep into the wall as to weaken the bearing. The

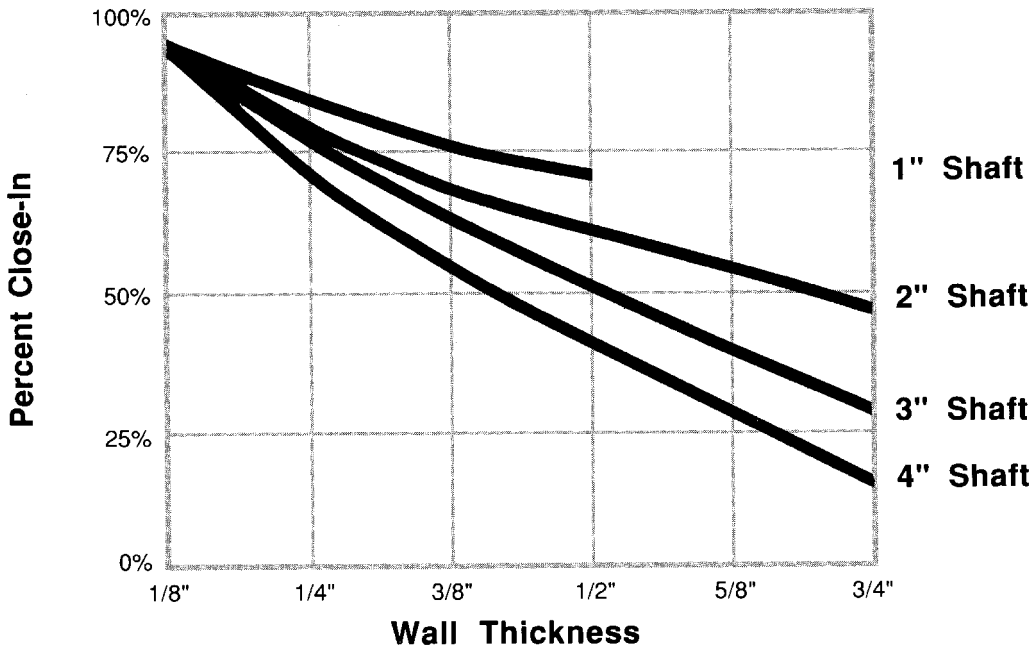


FIG. C-13 Close-in of inside diameters (approximation due to press or shrink fit). (Source: Advance Carbon Products.)

minimum wall thickness should be $\frac{1}{8}$ in, with $\frac{3}{16}$ in preferred. The recommended wall thickness is $\frac{1}{8}$ in + 10 percent of the bearing inside diameter. See Fig. C-13.

10. Tolerances. The product bearings can be machined to very close tolerances. The recommended tolerances are listed below, though closer tolerances can be machined at additional cost.

OD: ±0.001 in
ID: ±0.001 in
Length: ±0.010 in
Concentricity: 0.002 in TIR

Application example

Lube oil injection pump:

235°F operating temperature
316 stainless steel housing
316 stainless steel shaft, hard faced 0.020 deep, ground and polished
1770 rpm
3.092/3.090 housing diameter
2.4375 shaft diameter (maximum)

A. Material recommendation is AC-52, good up to 500°F.
B. Calculate expansion, close-in, and running clearance:

1. Housing: 316SS coefficient of thermal expansion is 8.3×10^{-6} in/in °F; @ 235°F, stainless steel housing expansion = (235–68°F) (3.092 dia.) (8.3×10^{-6} in/in °F) = 0.0043 in housing expansion

2. Bearing: AC-50 coefficient of thermal expansion is 2.6×10^{-6} in/in °F;
   @ 235°F, carbon OD expansion =
   (235–68°F) (3.092 dia.) (2.6×10^{-6} in/in °F) = 0.0010 in bearing expansion
3. Minimum interference fit =
   0.0015 in/in of diameter
   ×3.092 in OD
   0.0046 in min. OD Interference
4. Calculate OD:

	+0.0043	Housing expansion
	–0.0010	Bearing expansion
=	+0.0033	Expansion difference
add	+0.0046	Minimum interference fit
=	+0.0079	Total addition to OD
add	+3.0920	Housing ID
=	+3.0999	Bushing OD, minimum

   Since the housing has a 0.002 in tolerance, the tolerance for the bearing should be reduced to 0.001 in, so as to prevent a buildup of excessive tolerances. Final bushing OD = 3.100, +0.001, –0.000.
5. Close-in of ID at room temperature, due to press fit:
   Calculate minimum and maximum interference fits:

	Minimum:	
	3.100	Minimum bearing diameter
	–3.092	Maximum housing diameter
=	0.008	Minimum interference
	×80%	Close-in percentage
=	0.0064	Minimum close-in
	Maximum:	
	3.101	Maximum bearing diameter
	–3.090	Minimum housing diameter
=	0.011	Maximum interference
	×80%	Close-in percentage
=	0.0088	Maximum close-in

   0.0088/0.0064 = 0.0076 average close-in
6. Running clearance:
   2.4375 in shaft diameter
   ×0.002 in clearance/in of shaft diameter
   = 0.0049 minimum running clearance
7. Calculate bearing ID:
   Bearing close-in = +0.0076 in
   Running clearance = +0.0049 in
   High-limit shaft diameter = +2.4375 in
   Inside diameter = +2.450 in (tolerance: ±0.001)
8. Bearing length:
   Shaft diameter 2.4375 in
   ×1.5 ratio
   3.656 in preferred length

Seals

Mechanical seals are custom machined to a specification. *Note:* Material specifications stated here are typical values, and will vary with the size of the material.

Products that can be made with carbon-based materials:

Brushes: motor/generator

- Carbon
- Electrographitic
- Metal-graphite
- Graphite

Brush holders

Brush seater and cleaner

Brush tension scales

Carbon specialties

- Seals
- Steam turbine seals
- Gas turbine seals
- Hydraulic turbine seals
- Labyrinth seals
- Spherical seals
- Compressor seals
- Pump and compressor blades
- Bearings
- Thrust bearings
- Sliding guides
- Porous carbon filters

Commutator dressing and finishing stones

Electrical contacts

- Graphite
- Silver-graphite
- Copper-graphite

Metal impregnated carbon—graphite

Carbon Dioxide (CO_2); CO_2 Disposal

Carbon dioxide, an inert gas, is a byproduct of combustion. Large volumes of CO_2 result from combustion of fossil fuels. Industrial users of fossil fuels include gas and steam turbines. Industrial activity has contributed increasing volumes of CO_2 to the atmosphere. CO_2 has been found to be a greenhouse gas. Greenhouse gases contribute to the phenomenon of global warming. There have been changes in atmospheric CO_2 content that have "corrected themselves" in planetary history, but those changes occurred over millions of years versus the current trend established over just a few centuries. Technologies are now being developed to remove CO_2 from atmospheric solution to lessen the amount of CO_2 released into the atmosphere.

Due to chemical composition (number of atoms of carbon in a molecule of fuel), some fuels produce less CO_2 on a unit-weight basis than others that may be more commonly accepted on the market. This is one of the best ways of mitigating emissions of this greenhouse gas.

Research project activity in the field of CO_2 mitigation includes liquefication of CO_2 experiments (liquid CO_2 can be used in dry cleaning). There may eventually

also be an industrial grade process to solidify CO_2 by "turning" it into limestone or calcium carbonate. What follows is a description of an industrial process developed in Sweden, where CO_2 resulting from fossil fuel combustion is reinjected into the ground.

Technology and Cost Options for Capture and Disposal of Carbon Dioxide from Gas Turbines: A System Study for Swedish Conditions*

The current massive dependency on fossil fuels—90 percent of the world population's commercial production and consumption of energy—together with predictions of a considerable increase in the total world energy consumption during the coming decades, implies that the emissions of carbon dioxide from human activities will rise significantly over that period. Carbon dioxide (CO_2) is the largest anthropogenic contributor to the greenhouse effect. There is a broad consensus among scientists that the current and increased CO_2 emissions will increase the global mean temperature and affect local climates significantly, with numerous and far-reaching economic and environmental consequences.

Among several options for limiting future CO_2 emissions, capture and disposal of CO_2 from combustion gases has been studied within the IEA Greenhouse Gas Implementing Agreement. The process components of CO_2 capture have been demonstrated, and a complete demonstration plant (200 ton CO_2 per day recovered from boiler flue gases) is in operation at Shady Point in the United States (formerly ABB Lummus Crest; as of 2000, ABB is part of the Alstom Corporation). Disposal of CO_2 into sandstone aquifers is now under demonstration on a commercial scale. Since 1996, Statoil injects 1 million ton CO_2 per year into the Utsira sandstone formation at the Sleipner natural gas field.

Financed by NUTEK (The Swedish National Board for Industrial and Technical Development), a system study has been performed with the objective to assess how recent knowledge on the technical and economic options for the capture and the disposal of CO_2 from combustion gases could be implemented into the Swedish energy system.

Aquifers suitable for disposal of carbon dioxide

Surveys of earlier geologic investigations have indicated that geologic formations—aquifers—that should be suitable for CO_2 disposal exist in the south of Sweden–Denmark (South West Skåne and the eastern part of Zealand) and in the Baltic Sea between Gotland and Lithuania. The aquifer in Skåne-Denmark has the most favorable location with an estimated storage capacity of up to 10 Gton CO_2 of which the part in Skåne is estimated to have a storage capacity of up to 3.5 Gton. This could be compared to the yearly Swedish CO_2 emissions, approximately 60 Mton in 1995.

Capture of carbon dioxide from gas turbine–based power plants

Large-scale electric power production (500 MW power) with CO_2 capture has been studied for natural gas combined cycle (NGCC) and coal-based integrated gasification combined cycle (IGCC) within the IEA Greenhouse Gas R&D Programme. Based on these studies, we have studied the possibilities of recovering

* Source: Vattenfall Utveckling AB, Sweden; also, this section is adapted from extracts from ASME paper 98-GT-443.

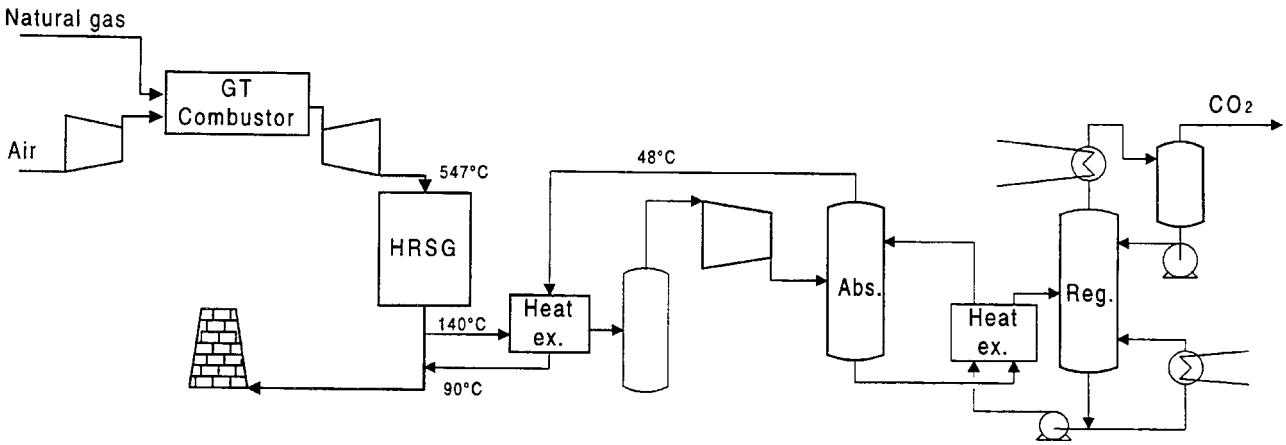


FIG. C-14 Process scheme for an NGCC (natural gas combined cycle) power plant with CO_2 capture. (Source: Vattenfall Utveckling AB.)

low-temperature heat from such processes as district heating. As a reference to the IGCC, performance and costs for pulverized coal combustion (PF, pulverized fuel) with and without CO_2 capture have also been estimated.

The process configurations, CO conversion rates, CO_2 removal efficiencies and other process parameters are the same as in IEA studies. Almost all process steps are based on proven technologies, and the process parameters have been chosen based on typically feasible designs and performances.

The NGCC plant consists of a single train of an advanced gas turbine. Like in some IEA studies, the calculations have been based on a Siemens 94.3A (turbine inlet temperature 1300°C and pressure ratio of 15–16) with a triple pressure reheat steam cycle (106 bar/30 bar/4.5 bar) (Fig. C-14). Due to the low partial pressure in the gas turbine exhaust gas, a CO_2 removal process based on chemical absorption, using a solvent such as MEA (MonoEthanol Amine), will be required. The assumed CO_2 removal efficiency is 85 percent. Regeneration of the solvent is performed by reboiling and stripping. Low-pressure steam for the regeneration is extracted from the steam cycle.

In the IGCC plant, coal is gasified at a high pressure (about 50 bar) and temperature (about 1400°C) with oxygen and steam in an entrained flow gasifier (Fig. C-15). In an IGCC plant without CO_2 capture, the fuel gas would be cooled and contaminants, such as dust and hydrogen sulfide, would be removed before burning the gas in the gas turbine combustor.

Due to its higher pressure and lower gas flow, it is advantageous to capture CO_2 in the fuel gas upstream of the gas turbine instead of from the exhaust gases. The fuel gas contains about 40 vol% CO, 28 vol% H_2, 18% H_2O, and 10 vol% CO_2. Since the CO (carbon monoxide) in the fuel gas would be emitted as CO_2 in the gas turbine exhaust gas, it must be converted to CO_2 prior to the CO_2 removal. This is achieved with steam according to the shift reaction, $CO + H_2O \Leftrightarrow CO_2 + H_2$. Medium pressure steam is extracted from the steam cycle. The steam demand is 0.5 kg H_2O/kg gas. The shift takes place in multiple catalytic reactors with intercooling at about 250–350°C.

After the sift, the CO_2 content in the gas has increased to 30 vol%. Because of the high pressure and concentration, a CO_2 removal process based on physical absorption, like the Selexol process, is most suitable for this application. The

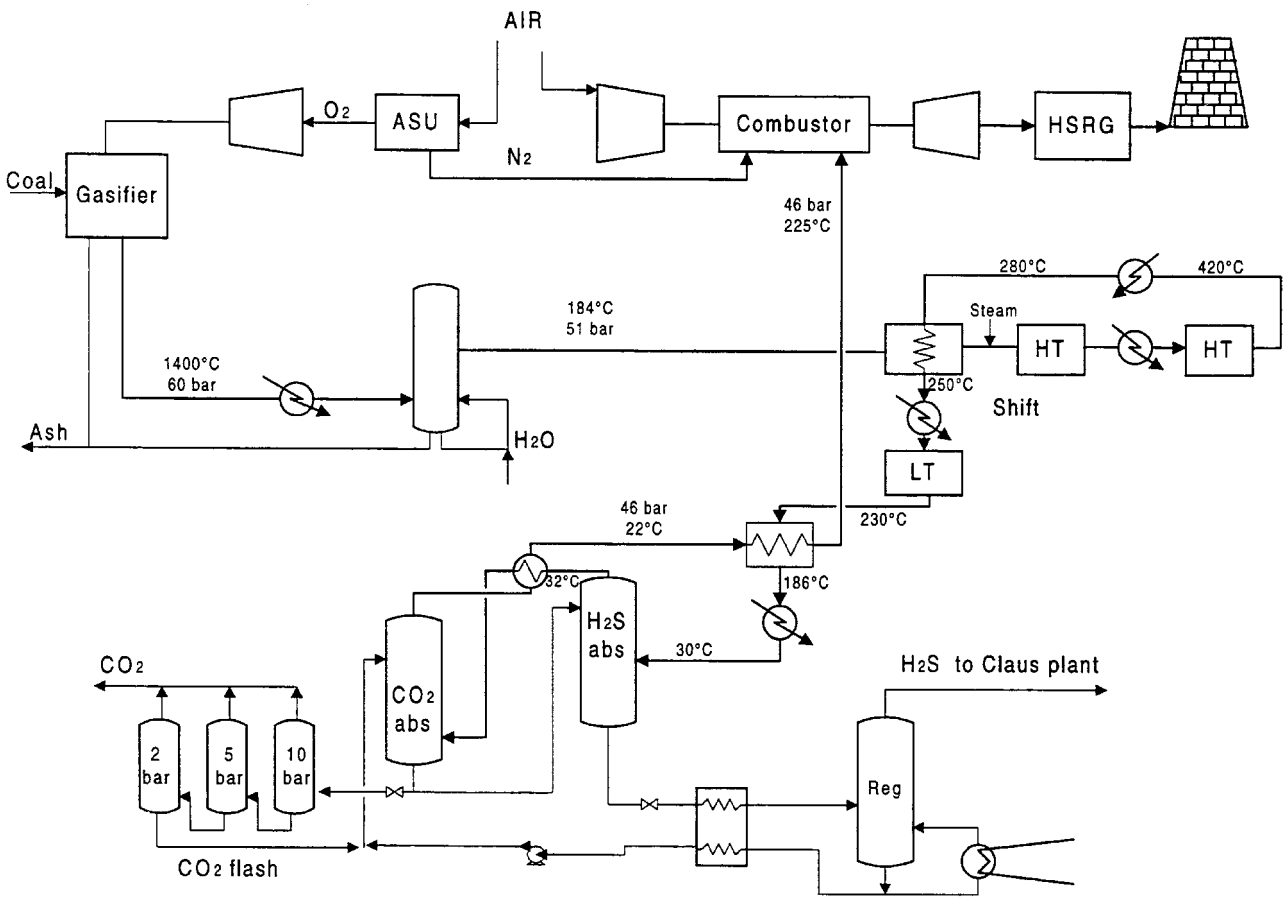


FIG. C-15 Process scheme for a coal-based IGCC (integrated gasification combined cycle) power plant with shift (conversion of CO to CO_2) and CO_2 capture upstream of the gas turbine combustion chamber. (Source: Vattenfall Utveckling AB.)

removal efficiency is assumed to be 90 percent. Hydrogen sulfide is selectively removed before the CO_2 removal. The sulfur-rich gas is transferred to a Claus unit, where elementary sulfur is produced. Regeneration of the absorbent is achieved by temperature increase and flashing. Low-pressure steam for the regeneration is extracted from the steam cycle. The dry isolated CO_2 is pressurized and liquefied.

After CO_2 removal, the hydrogen-rich fuel gas is burned in a gas turbine. A hydrogen-rich gas would most likely be a good gas turbine fuel. The gas turbine combustor must, of course, be designed for this type of fuel gas, since hydrogen has somewhat different combustion characteristics than natural gas. Combustion of hydrogen/steam mixtures for utilization in future advanced gas turbine cycles is investigated by Westinghouse. As in an IEA study, a Siemens V94.4 gas turbine has been assumed.

Carbon dioxide neutral coproduction of methanol, power, and district heating

Carbon dioxide neutral production and utilization of methanol as an automotive fuel for the transport sector integrated with production of electric power and district heat could be achieved with biomass combined with natural gas or coal as a raw material. An amount of CO_2 corresponding to the carbon in the fossil fuel then has to be captured and disposed into, e.g., an aquifer. Examples of a few such options

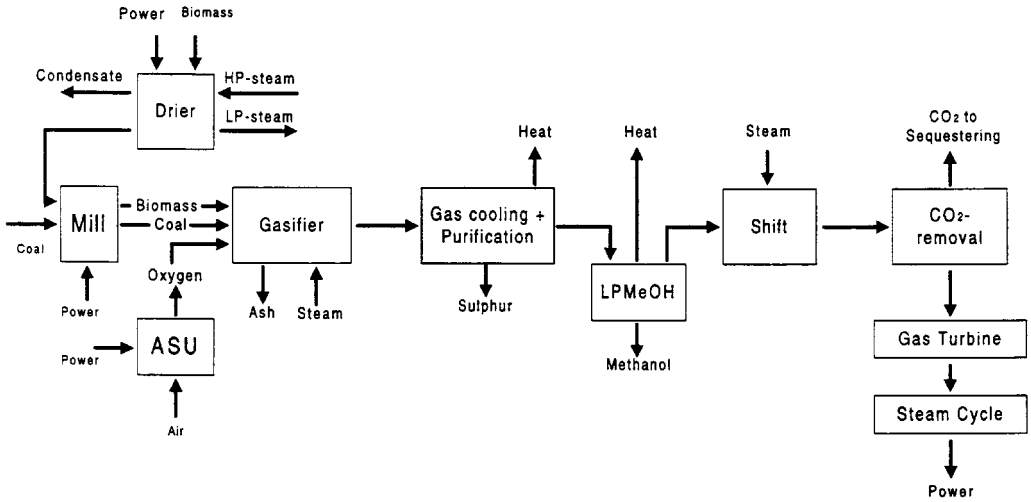


FIG. C-16 CO_2 neutral production of methanol, power and district heat by co-gasification of biomass and coal combined with CO_2 capture. (Source: Vattenfall Utveckling AB.)

have been studied based on IEA studies, other literature, and Vattenfall in-house information.

Co-gasification of biomass and coal. Coal and biomass are gasified in an entrained flow gasifier at 1400°C, 40 bar with oxygen and steam (Fig. C-16). Before being gasified, the biomass is dried in a steam drier, lowering its moisture content from about 50 percent to 10 percent, followed by milling. The air separation unit (ASU) is based on cryogenic separation. The syngas generated in the gasifier is cooled and cleaned from dust and sulfur. Heat is extracted to be used in the steam cycle.

The syngas contains about 28 vol% H_2 and 39 vol% CO at the inlet of the methanol synthesis reactor. Since both biomass and coal have low hydrogen contents, a novel methanol synthesis process under development by Chem. Systems/Air Products in the U.S., called LPMeOH (Liquid Phase Methanol Synthesis), has been selected. This process is less sensitive to the inlet syngas composition—mainly the ratio of $(H_2 - CO_2)/(CO + CO_2)$—than the current commercially available methanol synthesis processes. It has been assumed that 20 percent of the carbon input in the fuel is converted to methanol. Assuming only CO reacts according to $CO + 2H_2 \rightarrow CH_3OH$, the CO conversion is 25 percent on a molar basis. The methanol synthesis reaction is highly exothermic, and the released heat is utilized in the steam cycle.

The unreacted outlet gas from the methanol synthesis reactor contains mainly CO and H_2O. By adding steam, the CO is converted to CO_2 according to $CO + H_2O \Leftrightarrow H_2 + CO_2$ in the shift reactors. Like in an IGCC power plant case, a 95 percent conversion of CO has been assumed. The CO_2 content in the gas then increases from 14 vol% to 44 vol%. CO_2 is captured in a Selexol plant. The removal efficiency has been assumed to be 87 percent, which is close to the assumption for the IGCC power plant case.

After the CO_2 removal, the remaining gas rich in H_2 is burned in the gas turbine combustor. The gas turbine has been scaled to the actual fuel gas capacity from the Siemens V 94.4 gas turbine in the IGCC case, assuming unchanged performance. Heat from the gas turbine exhaust gases is utilized to generate steam for the bottoming cycle in a heat recovery boiler.

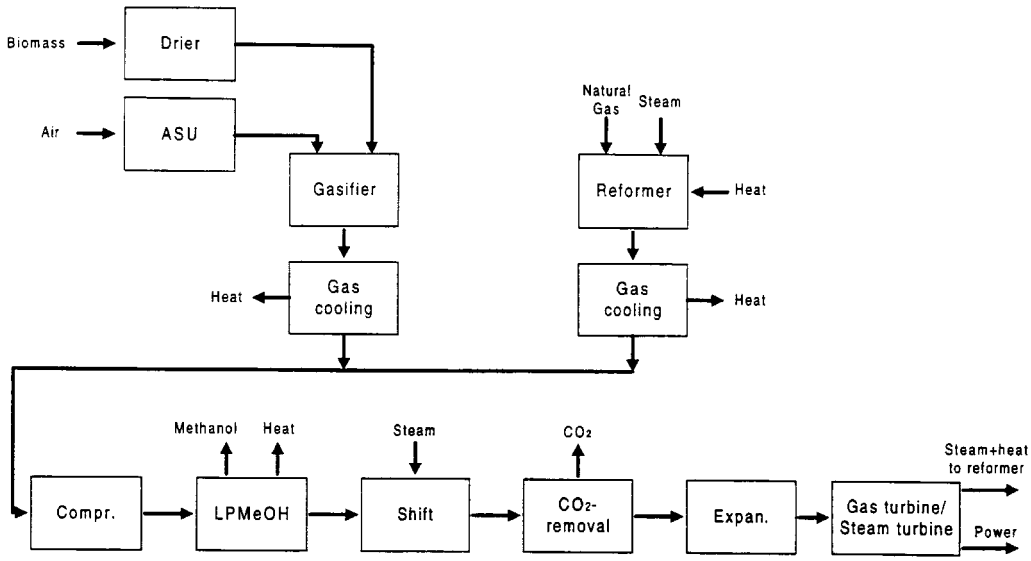


FIG. C-17 CO_2 neutral production of methanol, power, and district heat by gasification of biomass and reforming of natural gas in series combined with CO_2 capture. (Source: Vattenfall Utveckling AB.)

Gasification of biomass and reforming of natural gas in series. In this configuration, syngas is produced from biomass, oxygen, and steam in a fluidized bed gasifier at about 950°C, 20 bar. Before being gasified, the biomass is dried in a steam drier, lowering its moisture content from about 50 percent to 15–20 percent. The syngas from the gasifier is cleaned from dust using a ceramic filter at about 500°C and is then mixed with syngas from a natural gas reformer. Since both the gasifier and the reformer are operating at about 20 bar, the syngas mixture must be compressed before entering the methanol synthesis reactor. Since the gas composition is not optimal for a conventional methanol synthesis process, the LPMeOH process has been selected (Fig. C-17).

The unreacted outlet gas from the methanol synthesis reactor is shifted before the CO_2 removal. The CO_2 content in the gas then increases from 7 to about 18 vol%. Like in the IGCC power plant case, a 90 percent removal has been assumed. The remaining gas, rich in hydrogen (69 vol%), is expanded to the pressure required for the gas turbine combustor. The gas turbine has been scaled to the actual fuel gas capacity from the Siemens V 94.4 gas turbine in the IGCC case, assuming unchanged performance. The heat from the gas turbine exhaust gas is utilized in the steam cycle and for heating the reformer.

Energy efficiencies and costs when capturing carbon dioxide

The calculated efficiencies with and without CO_2 capture for the gas turbine–based power plants and for the described examples of CO_2 neutral coproduction of methanol, electric power, and district heat are summarized in Table C-2. Additional costs, due to the CO_2 capture, were estimated based on data from IEA studies, other literature, and this information source's in-house information. The results are summarized in Table C-2.

CO_2 capture and recovery consumes electricity and energy at high temperatures at the same time as energy at low temperatures can be recovered. This does not

TABLE C-2 Calculated Efficiencies and Carbon Dioxide Capture Costs for Electric Power Plants and for Carbon Dioxide Neutral Coproduction of Methanol, Electric Power, and District Heating

Capital costs:	7 percent real interest rate, 20 years economic lifetime
Fuel costs:	Natural gas 100 SEK/MWh, coal 50 SEK/MWh, biomass 120 SEK/MWh
District heat credit:	150 SEK/MWh
For Methanol and Electricity	
Methanol credit:	230 SEK/MWh (assumed world market price 1 SEK/liter)
Electric power credit:	280 SEK/MWh (calculated production cost from natural gas without CO_2 capture)

	Power Plant 6000 h/year			Methanol + Electricity 8000 h/year	
Fossil Fuel, MW (LHV)	Natural Gas 645	Coal IGCC 870	Biomass, MW (LHV) Fossil Fuel, MW (LHV)	385 Natural Gas 1245	385 Coal 870
			Methanol prod., MW	280	365
Net electricity, MW	300	320	Net electricity, MW	505	350
District heating, MW	55	85	District heating, MW	90	45
Net efficiencies			Net efficiencies		
Without CO_2 Capture			*With CO_2 Capture*		
Electric power, %	56	43	Methanol, %	17	29
With CO_2 Capture					
Electric power, %	47	37	Electric power, %	31	28
District heating, %	7	10	District heating, %	6	4
Total, %	54	47	Total, %	54	61
Captured CO_2, tons/h	120	260	Captured CO_2, tons/h	260	315
CO_2 Capture Costs			CO_2 Capture Costs		
SEK/ton CO_2	220	100	SEK/ton CO_2	340	145
SEK/MWh el	150	110 190 (IGCC with -PE without)	SEK/MWh (el + methanol)	110	65

mean a total energy loss for a power plant, but since a smaller fraction of the total energy will be available at higher temperatures (exergy loss), and a larger fraction will be available at lower temperatures, the electric efficiencies will be reduced. With the numerous district heating networks in Sweden, this means that it could be possible to compensate for the losses of electric efficiency by recovering energy at low temperatures as district heating. If sufficient quantities of low temperature heat can be sold, the total efficiencies will be nearly the same as for the corresponding power plants without capture of CO_2.

The calculated capture costs per metric ton CO_2 become higher for lower capture capacities than for higher capacities, which shows the scale economy for the capture and recovery process parts.

The estimated costs per ton CO_2 for the methanol and electricity cases are higher than for the power plants with similar capture capacities. This is due mainly to the choices of credits for methanol and electric power. Both these credits reflect natural gas–based production, which is less complex and therefore less costly than when solid fuels—biomass and/or coal—are used. At the same time, the capture costs per MWh (electricity + methanol) become about the same or lower as for the power plants. The main reason for this is that substantial fractions of the total fuel inputs

are biomass and only the CO_2 corresponding to the carbon in the fossil fuel input is captured. The quantities of captured CO_2 then become smaller in relation to the quantities of methanol plus electricity than the quantities of captured CO_2 in relation to the quantities of electricity produced in the power plants.

The most economic power production process for coal without CO_2 capture is PF (pulverized coal combustion) with a supercritical steam cycle. Its electric efficiency is today almost as high as the efficiency for an IGCC. With CO_2 capture, the steam consumption, auxiliary power need and cost for CO_2 capture from a boiler flue gas with low CO_2 concentration (requires chemical absorption) are higher than for CO_2 capture from a pressurized fuel gas with high CO_2 concentration (physical absorption will be sufficient). Consequently, the loss of electric power efficiency and the cost increase for CO_2 capture become substantially lower for an IGCC than for a PF. This results in IGCC being the most economic coal-based power process with CO_2 capture, also achieving the highest electric efficiency.

Transport and disposal of carbon dioxide

Captured CO_2 could be transported into an aquifer using a pipeline, ship, or possibly a train. A pipeline is more economic than ship (or train) for shorter distances, while ship transport seems to be the most feasible option for longer distances. However, for larger CO_2 quantities, pipelines may be the most economic option also for longer distances. The transport costs per ton CO_2 vary due to the distance and the quantities. To give an idea of these costs, estimated transport costs for two cases, a 40-km pipeline and a 700-km ship transport are shown in Table C-3.

The costs for injection of CO_2 into an aquifer have been estimated to 60–70 SEK/ton CO_2 ("SEK" is Swedish kroner) for a few studied examples based mainly on data from Danish studies on aquifer disposal.

Examples of total costs for capture, transport, and disposal of carbon dioxide

The estimated total cost for capture and recovery, transport, and injection into an aquifer of CO_2 consequently vary depending on energy production process, transport distance, and the scale—especially for the CO_2 transport. This is illustrated in Table C-4.

TABLE C-3 Estimated Examples of Transport Costs for Carbon Dioxide (Ekström, 1997)

	Pipeline	Ship
Distance, km	40	700
Quantity of transported CO_2, tons/h	120	90
Transport cost, SEK/ton CO_2	20	110

TABLE C-4 Summary of Estimated Costs for the Capture, Transport, and Disposal of Carbon Dioxide

	Power Plants 300–320 MW el 6000 h/year	Methanol and Electricity Plants 715–785 MW (el + methanol) 8000 h/year
Captured CO_2, tons/h	120–260	260–315
Capture and Recovery of CO_2, SEK/ton CO_2	100–220	145–340
Transport to Aquifer (40–700 km), SEK/ton CO_2	20–110	
Injection into Aquifer, SEK/ton CO_2	60–70	

For most of the studied cases, the estimated total costs are in the same order of magnitude as the current carbon dioxide tax in Sweden (365 SEK/ton CO_2).

If the distance to a suitable aquifer is long, the CO_2 transport costs can be significantly reduced, by collecting the captured CO_2 from more than one energy-producing plant into one high-capacity pipeline, because of the favorable economy of scale for pipeline transport.

Conclusions

Conclusions from the study are:

- Capture and disposal of CO_2 may be considered as one of the opportunities to reduce the total Swedish CO_2 emissions.
- The total estimated costs per ton CO_2 captured and disposed are, with the assumptions used, of the same order of magnitude as the current Swedish CO_2 taxes. Plant owners will have to be credited for the captured and disposed CO_2 in order to make this option economically justifiable and interesting for them.
- The process steps for CO_2 capture and recovery are well-proven and large-scale natural gas or coal-fired gas turbine–based power plants with CO_2 capture could be constructed using commercially available techniques.
- Pipeline transport of CO_2 is practiced. Large-scale ship or train transport systems are not developed for CO_2.
- Injection and disposal of CO_2 into aquifers is now at full-scale operation in the North Sea.
- The costs for CO_2 capture vary depending mainly on energy conversion process and scale. Ongoing process developments are likely to improve the economy.
- It will be important for the total economy to find favorable combinations of energy conversion, CO_2 capture and recovery, transport, and disposal. There is also a need to reduce uncertainties in the available basis for estimation of costs for large-scale transport, injection, and disposal of CO_2 into aquifers.
- For power plants, the losses of electric power efficiencies due to CO_2 capture could be compensated for in terms of maintaining total energy efficiencies, if sufficient quantities of district heat could be sold. The heat quantities produced for the plant capacities chosen in this study are likely to be suitable for a reasonable number of Swedish district heating networks.
- The CO_2 emissions from the transport sector could be reduced by combining CO_2 capture with the production of automotive fuel (methanol), electric power, and heat from biomass combined with natural gas or coal. Optimization work is needed to find the most favorable process configurations.
- Favorably localized aquifers that should be suitable for CO_2 disposal and with a substantial storing capacity exist in South West Skåne-Denmark (eastern part of Zealand).
- Several aspects need to be clarified regarding aquifer conditions, disposal, and large-scale CO_2 transport in order to make more accurate assessments. Swedish–Norwegian–Danish cooperation projects regarding further clarification of techniques and geology for aquifer disposal and large-scale CO_2 transport systems would be beneficial.
- A Swedish implementation of capture and disposal of CO_2 from combustion of fossil fuels will require political decisions and development of legal frameworks. The investigations of these aspects were outside the scope of this study.

Castings (*see* Metallurgy)

Cells

Chemical cells are a specialized, often custom-designed item, common in applications such as metallurgical processes, for example, electroplating. *See* Some Commonly Used Specifications, Codes, Standards, and Texts.

Cement; Portland Cement

Cement is made by several companies around the world to exact standard specifications. When mixed with varying grades of air, aggregate, and soil, concrete of varying strengths is produced. Extracts from American Society for Testing and Materials (ASTM) specification C 150-95 for Portland cement follow.

The manufacture of cement raises significant environmental concerns, and standards have been developed around these considerations. Extracts from a paper that describes cement manufacture in Canada and Canada's emission guidelines with respect to cement manufacturing kilns, also follows. Specifications will vary slightly from country to country. Legislation, such as U.S.–Canada Air Agreements and global conventions on issues such as air quality and global warming (e.g., Kyoto, Rio, Montreal) are helping introduce communality into specifications of products that affect air emissions, such as cement.

Extracts from *Standard Specification for Portland Cement*, ASTM C 150-95[1]

This standard is issued under the fixed designation C 150; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the Department of Defense. Consult the DoD Index of Specifications and Standards for the specific year of issue which has been adopted by the Department of Defense.

1. Scope

1.1 This specification covers eight types of Portland cement, as follows (see Note 1 in section 4 below):

1.1.1 *Type I*—For use when the special properties specified for any other type are not required.

1.1.2 *Type IA*—Air-entraining cement for the same uses as Type I, where air-entrainment is desired.

1.1.3 *Type II*—For general use, more especially when moderate sulfate resistance or moderate heat of hydration is desired.

1.1.4 *Type IIA*—Air-entraining cement for the same uses as Type II, where air-entrainment is desired.

1.1.5 *Type III*—For use when high early strength is desired.

1.1.6 *Type IIIA*—Air-entraining cement for the same use as Type III, where air-entrainment is desired.

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[1] This specification is under the jurisdiction of ASTM Committee C-1 on Cement and is the direct responsibility of Subcommittee C01.10 on Portland Cement. Current edition approved June 15, 1995. Published August 1995. Originally published as C 150-40 T. Last previous edition C 150-94b.

1.1.7 *Type IV*—For use when a low heat of hydration is desired.

1.1.8 *Type V*—For use when high sulfate resistance is desired.

1.2 When both SI and inch-pound units are present, the SI units are the standard. The inch-pound units are approximations listed for information only.

2. Referenced documents

2.1 *ASTM Standards*:

C 33 Specification for Concrete Aggregates[2]

C 109 Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or 50-mm Cube Specimens)[3]

C 114 Test Methods for Chemical Analysis of Hydraulic Cement[3]

C 115 Test Method for Fineness of Portland Cement by the Turbidimeter[3]

C 151 Test Method for Autoclave Expansion of Portland Cement[3]

C 183 Practice for Sampling and the Amount of Testing of Hydraulic Cement[3]

C 185 Test Method for Air Content of Hydraulic Cement Mortar[3]

C 186 Test Method for Heat of Hydration of Hydraulic Cement[3]

C 191 Test Method for Time of Setting of Hydraulic Cement by Vicat Needle[3]

C 204 Test Method for Fineness of Hydraulic Cement by Air Permeability Apparatus[3]

C 226 Specification for Air-Entraining Additions for Use in the Manufacture of Air-Entraining Portland Cement[3]

C 266 Test Method for Time of Setting of Hydraulic Cement Paste by Gillmore Needles[3]

C 451 Test Method for Early Stiffening of Portland Cement (Paste Method)[3]

C 452 Test Method for Potential Expansion of Portland Cement Mortars Exposed to Sulfate[3]

C 465 Specification for Processing Additions for Use in the Manufacture of Hydraulic Cements[3]

C 563 Test Method for Optimum SO_3 in Portland Cement[3]

C 1038 Test Method for Expansion of Portland Cement Mortar Bars Stored in Water[3]

3. Terminology

3.1 *Definitions*:

3.1.1 *Portland cement*—a hydraulic cement produced by pulverizing clinker consisting essentially of hydraulic calcium silicates usually containing one or more of the forms of calcium sulfate as an interground addition.

3.1.2 *Air-entraining portland cement*—a hydraulic cement produced by pulverizing clinker consisting essentially of hydraulic calcium silicates, usually containing one or more of the forms of calcium sulfate as an interground addition, and with which there has been interground an air-entraining addition.

4. Ordering information

4.1 Orders for material under this specification shall include the following:

4.1.1 This specification number and date,

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[2] *Annual Book of ASTM Standards*, Vol. 04.02.

[3] *Annual Book of ASTM Standards*, Vol. 04.01.

4.1.2 Type or types allowable. If no type is specified, Type I shall be supplied,

4.1.3 Any optional chemical requirements from Table C-6, if desired,

4.1.4 Type of setting-time test required, Vicat or Gillmore. If not specified, the Vicar shall be used,

4.1.5 Any optional physical requirements from Table C-8, if desired.

NOTE 1—Attention is called to the fact that cements conforming to the requirements for all types may not be carried in stock in some areas. In advance of specifying the use of other than Type I cement, it should be determined whether the proposed type of cement is or can be made available.

5. Additions

5.1 The cement covered by this specification shall contain no addition except as follows:

5.1.1 Water or calcium sulfate, or both, may be added in amounts such that the limits shown in Table C-5 for sulfur trioxide and loss-on-ignition shall not be exceeded.

5.1.2 At the option of the manufacturer, processing additions may be used in the manufacture of the cement, provided such materials in the amounts used have been shown to meet the requirements of Specification C 465.

5.1.3 Air-entraining portland cement shall contain an interground addition conforming to the requirements of Specification C 226.

6. Chemical composition

6.1 Portland cement of each of the eight types shown in section 1 above shall conform to the respective standard chemical requirements prescribed in Table C-5. In addition, optional chemical requirements are shown in Table C-6.

7. Physical properties

7.1 Portland cement of each of the eight types shown in section 1 above shall conform to the respective standard physical requirements prescribed in Table C-7. In addition, optional physical requirements are shown in Table C-8.

8. Sampling

8.1 When the purchaser desires that the cement be sampled and tested to verify compliance with this specification, sampling and testing should be performed in accordance with Practice C 183.

8.2 Practice C 183 is not designed for manufacturing quality control and is not required for manufacturer's certification.

9. Test methods

9.1 Determine the applicable properties enumerated in this specification in accordance with the following test methods:

9.1.1 *Air Content of Mortar*—Test Method C 185.

9.1.2 *Chemical Analysis*—Test Methods C 114.

9.1.3 *Strength*—Test Method C 109.

9.1.4 *False Set*—Test Method C 451.

9.1.5 *Fineness by Air Permeability*—Test Method C 204.

9.1.6 *Fineness by Turbidimeter*—Test Method C 115.

9.1.7 *Heat of Hydration*—Test Method C 186.

TABLE C-5 Standard Chemical Requirements

Cement Type[A]	I and IA	II and IIA	III and IIIA	IV	V
Silicon dioxide (SiO_2), min, %	—	20.0	—	—	—
Aluminum oxide (Al_2O_3), max, %	—	6.0	—	—	—
Ferric oxide (Fe_2O_3), max, %	—	6.0	—	6.5	—
Magnesium oxide (MgO), max, %	6.0	6.0	6.0	6.0	6.0
Sulfur trioxide (SO_3),[B] max, %					
When (C_3A)[C] is 8% or less	3.0	3.0	3.5	2.3	2.3
When (C_3A)[C] is more than 8%	3.5	[D]	4.5	[D]	[D]
Loss on ignition, max, %	3.0	3.0	3.0	2.5	3.0
Insoluble residue, max, %	0.75	0.75	0.75	0.75	0.75
Tricalcium silicate (C_3S)[C] max, %	—	—	—	35[F]	—
Dicalcium silicate (C_2S)[C] min, %	—	—	—	40[E]	—
Tricalicum aluminate (C_3A)[C] max, %	—	8	15	7[E]	5[F]
Tetracalcium aluminoferrite plus twice the tricalcium aluminate[C] ($C_4AF + 2(C_3A)$), or solid solution ($C_4AF + C_2F$), as applicable, max, %	—	—	—	—	25[F]

[A] See Note 1 in section 4.

[B] There are cases where optimum SO_3 (using Test Method C 553) for a particular cement is close to or in excess of the limit in this specification. In such cases where properties of a cement can be improved by exceeding the SO_3 limits stated in this table, it is permissible to exceed the values in the table, provided it has been demonstrated by Test Method C 1038 that the cement with the increased SO_3 will not develop expansion in water exceeding 0.020% at 14 days. When the manufacturer supplies cement under this provision, he shall, upon request, supply supporting data to the purchaser.

[C] The expressing of chemical limitations by means of calculated assumed compounds does not necessarily mean that the oxides are actually or entirely present as such compounds.

When expressing compounds, C – CaO, S – SiO_2, A – Al_2O_3, F – Fe_2O_3. For example, C_3A – $3CaO{\cdot}Al_2O_3$.

Titanium dioxide and phosphorus pentoxide (TiO_2 and P_2O_5) shall be included with the Al_2O_3 content. The value historically and traditionally used for Al_2O_3 in calculating potential compounds for specification purposes is the ammonium hydroxide group minus ferric oxide ($R_2O_3 - Fe_2O_3$) as obtained by classical wet chemical methods. This procedure includes as Al_2O_3 the TiO_2, P_2O_5 and other trace oxides which precipitate with the ammonium hydroxide group in the classical wet chemical methods. Many modern instrumental methods of cement analysis determine aluminum or aluminum oxide directly without the minor and trace oxides included by the classical method. Consequently, for consistency and to provide comparability with historic data and among various analytical methods, when calculating potential compounds for specification purposes, those using methods which determine Al or Al_2O_3 directly should add to the determined Al_2O_3 weight quantities of P_2O_5. TiO_2 and any other oxide except Fe_2O_3 which would precipitate with the ammonium hydroxide group when analyzed by the classical method and which is present in an amount of 0.05 weight % or greater. The weight percent of minor or trace oxides to be added to Al_2O_3 by those using direct methods may be obtained by actual analysis of those oxides in the sample being tested or estimated from historical data on those oxides on cements from the same source, provided that the estimated values are identified as such.

When the ratio of percentages of aluminum oxide to ferric oxide is 0.64 or more, the percentages of tricalcium silicate, dicalcium silicate, tricalcium aluminate, and tetracalcium aluminoferrite shall be calculated from the chemical analysis as follows:

$$\text{Tricalcium silicate} = (4.071 \times \%\ CaO) - (7.600 \times \%\ SiO_2) - (6.718 \times \%\ Al_2O_3) - (1.430 \times \%\ Fe_2O_3) - (2.852 \times \%\ SO_3)$$
$$\text{Dicalcium silicate} = (2.867 \times \%\ SiO_2) - (0.7544 \times \%\ C_3S)$$
$$\text{Tricalcium aluminate} = (2.650 \times \%\ Al_2O_3) - (1.692 \times \%\ Fe_2O_3)$$
$$\text{Tetracalcium aluminoferrite} = 3.043 \times \%\ Fe_2O_3$$

When the alumina-ferric oxide ratio is less than 0.64, a calcium aluminoferrite solid solution (expressed as ss($C_4AF + C_2F$)) is formed. Contents of this solid solution and of tricalcium silicate shall be calculated by the following formulas:

$$ss(C_4AF + C_2F) = (2.100 \times \%\ Al_2O_3) + (1.702 \times \%\ Fe_2O_3)$$
$$\text{Tricalcium silicate} = (4.071 \times \%\ CaO) - (7.600 \times \%\ SiO_2) - (4.479 \times \%\ Al_2O_3) - (2.859 \times \%\ Fe_2O_3) - (2.852 \times \%\ SO_3).$$

No tricalcium aluminate will be present in cements of this composition. Dicalcium silicate shall be calculated as previously shown.

In the calculation of all compounds the oxides determined to the nearest 0.1% shall be used.

All values calculated as described in this note shall be reported to the nearest 1%.

[D] Not applicable.

[E] Does not apply when the heat of hydration limit in Table C-8 is specified.

[F] Does not apply when the sulfate resistance limit in Table C-8 specified.

9.1.8 *Autoclave Expansion*—Test Method C 151.

9.1.9 *Time of Setting by Gillmore Needles*—Test Method C 266.

9.1.10 *Time of Setting by Vicat Needles*—Test Method C 191.

9.1.11 *Sulfate Resistance*—Test Method C 452 (sulfate expansion).

9.1.12 *Calcium Sulfate (expansion of) Mortar*—Test Method C 1038.

9.1.13 *Optimum SO_3*—Test Method C 563.

TABLE C-6 Optional Chemical Requirements[A]

Cement Type	I and IA	II and IIA	III and IIIA	IV	V	Remarks
Tricalcium aluminate (C_3A),[B] max, %	—	—	8	—	—	For moderate sulfate resistance
Tricalcium aluminate (C_3A),[B] max, %	—	—	5	—	—	For high sulfate resistance
Sum of tricalcium silicate and tricalcium aluminate,[B] max, %	—	58[C]	—	—	—	For moderate heat of hydration
Equivalent alkalies ($Na_2O + 0.658K_2O$), max, %	0.60[D]	0.60[D]	0.60[D]	0.60[D]	0.60[D]	Low-alkali cement

[A] These optional requirements apply only if specifically requested. Availability should be verified. See Note 1 in section 4.

[B] The expressing of chemical limitations by means of calculated assumed compounds does not necessarily mean that the oxides are actually or entirely present as such compounds.

When expressing compounds, C = CaO, S = SiO_2, A = Al_2O_3, F = Fe_2O_3. For example, $C_3A = 3CaO{\cdot}Al_2O_3$.

Titanium dioxide and phosphorus pentoxide (TiO_2 and P_2O_5) shall be included with the Al_2O_3 content. The value historically and traditionally used for Al_2O_3 in calculating potential compounds for specification purposes is the ammonium hydroxide group minus ferric oxide ($R_2O_3 - Fe_2O_3$) as obtained by classical wet chemical methods. This procedure includes as Al_2O_3 the TiO_2, P_2O_5 and other trace oxides which precipitate with the ammonium hydroxide group in the classical wet chemical methods. Many modern instrumental methods of cement analysis determine aluminum or aluminum oxide directly without the minor and trace oxides included by the classical method. Consequently, for consistency and to provide comparability with historic data and among various analytical methods, when calculating potential compounds for specification purposes, those using methods which determine Al or Al_2O_3 directly should add to the determined Al_2O_3 weight quantities of P_2O_5. TiO_2 and any other oxide except Fe_2O_3 which would precipitate with the ammonium hydroxide group when analyzed by the classical method and which is present in an amount of 0.05 weight % or greater. The weight percent of minor or trace oxides to be added to Al_2O_3 by those using direct methods may be obtained by actual analysis of those oxides in the sample being tested or estimated from historical data on those oxides on cements from the same source, provided that the estimated values are identified as such.

When the ratio of percentages of aluminum oxide to ferric oxide is 0.64 or more, the percentages of tricalcium silicate, dicalcium silicate, tricalcium aluminate and tetracalcium aluminoferrite shall be calculated from the chemical analysis as follows:

$$\text{Tricalcium silicate} = (4.071 \times \%\ CaO) - (7.600 \times \%\ SiO_2) - (6.718 \times \%\ Al_2O_3) - (1.430 \times \%\ Fe_2O_3) - (2.852 \times \%\ SO_3)$$
$$\text{Dicalcium silicate} = (2.867 \times \%\ SiO_2) - (0.7544 \times \%\ C_3S)$$
$$\text{Tricalcium aluminate} = (2.650 \times \%\ Al_2O_3) - (1.692 \times \%\ Fe_2O_3)$$
$$\text{Tetracalcium aluminoferrite} = 3.043 \times \%\ Fe_2O_3$$

When the alumina-ferric oxide ratio is less than 0.64, a calcium aluminoferrite solid solution (expressed as $ss(C_4AF + C_2F)$) is formed. Contents of this solid solution and of tricalcium silicate shall be calculated by the following formulas:

$$ss(C_4AF + C_2F) = (2.100 \times \%\ Al_2O_3) + (1.702 \times \%\ Fe_2O_3)$$
$$\text{Tricalcium silicate} = (4.071 \times \%\ CaO) - (7.600 \times \%\ SiO_2) - (4.479 \times \%\ Al_2O_3) - (2.859 \times \%\ Fe_2O_3) - (2.852 \times \%\ SO_3).$$

No tricalcium aluminate will be present in cements of this composition. Dicalcium silicate shall be calculated as previously shown.

In the calculation of all compounds the oxides determined to the nearest 0.1% shall be used.

All values calculated as described in this note shall be reported to the nearest 1%.

[C] The optional limit for heat of hydration in Table C-8 shall not be requested when this optional limit is requested.

[D] This limit may be specified when the cement is to be used in concrete with aggregates that may be deleteriously reactive. Reference should be made to Specification C 33 for suitable criteria of deleterious reactivity.

10. Inspection

10.1 Inspection of the material shall be made as agreed upon by the purchaser and the seller as part of the purchase contract.

11. Rejection

11.1 The cement may be rejected if it fails to meet any of the requirements of this specification.

11.2 Cement remaining in bulk storage at the mill, prior to shipment, for more than 6 months, or cement in bags in local storage in the hands of a vendor for more than 3 months, after completion of tests, may be retested before use and may be rejected if it fails to conform to any of the requirements of this specification.

11.3 Packages shall identify the mass contained as net weight. Packages more than 2 percent below the weight marked thereon may be rejected; and if the average weight of packages in any shipment, as shown by weighing 50 packages taken at random, is less than that marked on the packages, the entire shipment may be rejected.

TABLE C-7 Standard Physical Requirements

Cement Type[A]	I	IA	II	IIA	III	IIIA	IV	V
Air content of mortar,[B] volume %:								
max	12	22	12	22	12	22	12	12
min	—	16	—	16	—	16	—	—
Fineness,[C] specific surface, m^2/kg (alternative methods):								
Turbidimeter test, min	160	160	160	160	—	—	160	160
Air permeability test, min	280	280	280	280	—	—	280	280
Autoclave expansion, max, %	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80
Strength, not less than the values shown for the ages indicated below:[D]								
Compressive strength, MPa (psi):								
1 day	—	—	—	—	12.0 (1,740)	10.0 (1,450)	—	—
3 days	12.0 (1,740)	10.0 (1,450)	10.0 (1,450) 7.0[F] (1,020)[F]	8.0 (1,160) 6.0[F] (870)[F]	24.0 (3,480)	19.0 (2,760)	—	8.0 (1,160)
7 days	19.0 (2,760)	16.0 (2,320)	17.0 (2,470) 12.0[F] (1,740)[F]	14.0 (2,030) 9.0[F] (1,310)[F]	—	—	7.0 (1,020)	15.0 (2,180)
28 days	—	—	—	—	—	—	17.0 (2,470)	21.0 (3,050)
Time of setting (alternative methods):[E]								
Gillmore test:								
Inital set, min, not less than	60	60	60	60	60	60	60	60
Final set, min, not more than	600	600	600	600	600	600	600	600
Vicat test:[G]								
Time of setting, min, not less than	45	45	45	45	45	45	45	45
Time of setting, min, not more than	375	375	375	375	375	375	375	375

[A] See Note 1 in section 4.
[B] Compliance with the requirements of this specification does not necessarily ensure that the desired air content will be obtained in concrete.
[C] Either of the two alternative fineness methods may be used at the option of the testing laboratory. However, when the sample falls to meet the requirements of the air-permeability test, the turbidimeter test shall be used, and the requirements in this table for the turbidimetric method shall govern.
[D] The strength at any specified test age shall be not less than that attained at any previous specified test age.
[E] The purchaser should specify the type of setting-time test required. In case he does not so specify, the requirements of the Vicat test only shall govern.
[F] When the optional heat of hydration or the chemical limit on the sum of the tricalcium silicate and tricalcium aluminate is specified.
[G] The time of setting is that described as initial setting time in Test Method C 191.

12. Manufacturer's statement

12.1 At the request of the purchaser, the manufacturer shall state in writing the nature, amount, and identity of the air-entraining agent used, and of any processing addition that may have been used, and also, if requested, shall supply test data showing compliance of such air-entraining addition with the provisions of Specification C 226, and of any such processing addition with Specification C 465.

13. Packaging and package marking

13.1 When the cement is delivered in packages, the words "Portland Cement," the type of cement, the name and brand of the manufacturer, and the mass of the cement contained therein shall be plainly marked on each package. When the cement is an air-entraining type, the words "air entraining" shall be plainly marked on each package. Similar information shall be provided in the shipping documents

accompanying the shipment of packaged or bulk cement. All packages shall be in good condition at the time of inspection.

NOTE 2—With the change to SI units, it is desirable to establish a standard SI package for portland cements. To that end 40 kg (88.18 lb) provides a convenient even-numbered mass reasonably similar to the traditional 94-lb (42.6384-kg) package.

14. Storage

14.1 The cement shall be stored in such a manner as to permit easy access for proper inspection and identification of each shipment, and in a suitable weather-tight building that will protect the cement from dampness and minimize warehouse set.

15. Manufacturer's certification

15.1 Upon request of the purchaser in the contract or order, a manufacturer's report shall be furnished at the time of shipment stating the results of tests made on samples of the material taken during production or transfer and certifying that the cement conforms to applicable requirements of this specification.

As previously noted, emissions guidelines for cement kilns vary from country to country. The sample country discussed in this case is Canada.

Extracts from "Development of CCME National Emission Guidelines for Cement Kilns"*

The guideline proposes emission limits for new kilns in the cement manufacturing industry, and makes some recommendations on emission reductions from existing plants that are being modified or upgraded. Regional or provincial regulatory authorities could decide to impose stricter standards in response to local air-quality problems.

The guideline was prepared through extensive consultation between industry, governments, and environmental groups. Principles that were considered to be important were those of pollution prevention, energy efficiency, cost-effectiveness, and a comprehensive view toward minimizing various emissions to reduce air pollution and greenhouse gases. A future guideline for lime kilns will also be developed subsequent to further research with the lime industry. Tables C-5 through C-7 describe characteristics of cement.

Cement kilns in the sample country (Canada)

Portland cement is mixed with sand, aggregates, and water to form the basic building material known as concrete. The production of cement, which comprises 10–15 percent of the final concrete mixture, is based on the conversion (pyroprocessing) of a mixture of limestone ($CaCO_3$) and shale or clay, into clinker material consisting of compounds of calcium oxide (CaO), by the addition of large quantities of heat in a coal- or gas-fired rotary kiln (Fig. C-18). The raw cement clinker exits the kiln and is usually mixed with 3 to 6 percent gypsum and then finely ground into powder. Sometimes it can also be mixed with flyash or other cementitious additives if blended cements are used. The characteristics of the raw feed materials, the finished product, and the resulting emissions are very site-

* Source: Environment Canada. Adapted with permission.

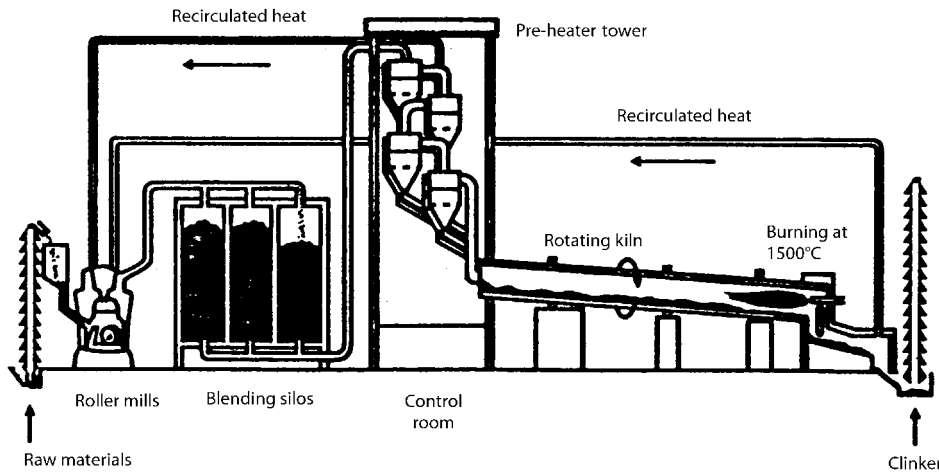


FIG. C-18 Basic schematic of cement production. (Source: Environment Canada.)

TABLE C-8 Optional Physical Requirements[A]

Cement Type	I	IA	II	IIA	III	IIIA	IV	V
False set, final penetration, min, %	50	50	50	50	50	50	50	50
Heat of hydration:								
7 days, max, kJ/kg (cal/g)	—	—	290 (70)[B]	290 (70)[B]	—	—	250 (80)[C]	—
28 days, max, kJ/kg (cal/g)	—	—	—	—	—	—	290 (70)[C]	—
Strength, not less than the values shown:								
Compressive strength, MPa (psi)								
28 days	28.0 (4,060)	22.0 (3,190)	28.0 (4,060) 22.0[B] (3,190)[B]	22.0 (3,190) 18.0[B] (2,610)[B]	—	—	—	—
Sulfate resistance,[D] 14 days, max, % expansion	—	—	—	—	—	—	—	0.040

[A] These optional requirements apply only if specifically requested. Availability should be verified. See Note 1 in section 4.
[B] The optional limit for the sum of the tricalcium silicate and tricalcium aluminate in Table C-6 shall not be requested when this optional limit is requested. These strength requirements apply when either heat of hydration or the sum of tricalcium silicate and tricalcium aluminate requirements are requested.
[C] When the heat of hydration limit is specified, it shall be instead of the limits of C3S, C2S, and C3A listed in Table C-5.
[D] When the sulfate resistance is specified, it shall be instead of the limits of C_3A and $C_4AF + 2C_3A$ listed in Table C-5.

specific depending upon the chemistry of the quarried limestone, the type of cement required, and the type of kiln used. See Tables C-5 through C-8.

The clinkering process begins with the feed of the raw limestone mixture into the higher end of the rotary kiln, where it is exposed to gradually increasing temperatures through an evaporation/preheat zone (100–400°C). In the middle of the kiln temperatures in the 500–900°C range cause calcination, liberating large amounts of carbon dioxide from the limestone. Near the lower burner end the remaining material is sintered at about 1500°C to form clinker pellets, which are subsequently cooled and sent to the grinding mill. Combustion gases flow in the opposite direction toward the stack located at the higher end of the kiln. NO_x formation is dominated by the thermal NO_x mechanism from combustion temperatures approaching 1900°C, in the pyroprocessing stage to sinter the material into a cement clinker, as well as a small quantity from initial limestone calcination.

Clinker production in an older style wet process long kiln, with a slurry raw material feed, uses chain sections at the feed end to contact the feed to evaporate the 30 percent water content. These kilns have high fuel consumption, about 6 GJ/tonne of clinker, as well as higher auxiliary electrical needs. Energy use was reduced by 15–20 percent with the advent of long dry-feed kilns in the mid-1960s, and then again with the use of suspension preheaters at the feed end. These tower preheaters use three to six stages of counterflow contact between the hot exhaust gases and the feed entering a much shorter kiln, resulting in a much improved 3.5 GJ/tonne energy consumption. The 1980s saw the implementation of the precalciner kilns, which are similar to the preheater type, but they move up to half of the fuel burning to the kiln inlet below the preheater tower. This produces calcination outside the kiln, allows for higher capacity shorter kilns, and also produces less NO_x emissions due to the staging aspects of fuel combustion.

The industry is comprised of 18 active cement plants across Canada, with 32 kilns in total, normally producing about ten million tonnes of cement. Kiln capacities range from the 500–1500 tonnes/day for wet and long dry kilns, to 2000–4500 tonnes/day for preheater/precalciner units. Although most of these kilns are of the older wet/long dry design, almost three-quarters of total production is now done by the ten large preheater/precalciner types. As production is shifted toward the new plants, and some of the older kilns such as the Richmond, BC, plant are converted to the higher-efficiency plant, average energy use will improve from 4.5 GJ/tonne to 3.4 GJ/tonne of clinker by year 2010. Fuel use consists mainly of coal and petroleum coke (60 percent), natural gas (25 percent), and oil- and waste-derived fuels (15 percent).

Atmospheric emissions from cement kilns

Air emissions from kilns arise because of the nature and chemistry of the raw material used and from the fuel burned in the kiln. Particulate emissions have traditionally been the main source of concern; PM emissions generally range from 0.3–1.0 kg/tonne from the combined kiln and clinker cooling facilities. Electrostatic precipitators are commonly employed; some humidification may be required to optimize moisture to maintain particle resistivity. Carbon monoxide levels in the ESP must be monitored to avoid explosive conditions. Some plants may also use fabric filter baghouses if flow conditions are appropriate and exhaust temperatures are not too high. Much of the collected kiln dust is often recirculated into the kiln feed.

NO_x emissions from cement kilns are primarily generated from the high combustion temperature at the main burner. They have a wide range of values, depending upon type of kiln and fuel used. See Tables C-9 and C-10. The average emission rate is about 3.7 kg_{NO_x} per tonne of clinker. Total NO_x emissions are in the range of 30–35 kilotonnes, depending on the production split between gas-fired plants (higher NO_x) in Western Canada, and mostly coal-fired plants in Eastern Canada. Table C-9 summarizes findings in the Radian study, based on averages from various research activities and from Canadian and international industry data. In similar kilns, coal firing produces less NO_x than natural gas, since thermal NO_x is the dominant mechanism. Note that this difference is less for precalciner kilns. Pyroprocessing does require high temperature and resulting NO_x emissions in the kiln firing zone, and emission monitoring has been used to optimize this clinker burning zone. Emission prevention potential from combustion can be realized by minimizing fuel use and by transferring the heat input to the feed end of a shorter kiln. There is limited experience with back-end NO_x emission control

TABLE C-9 Average NO_x Emissions from Cement Kilns (kg/tonne)

Fuel	Wet Kiln	Long Dry Kiln	Preheater	Precalciner
Gas	9	7–9.5	5.6	1.7–3
Coal	1.5–4	2.4–4.6	1.5–2.8	1.4–2

TABLE C-10 Comparison of NO_x Control Technologies for Cement Kilns

Control Technology	Technical Feasibility	Potential NO_x Reduction	Annualized Cost Range, 000$/yr	Cost Effectiveness, $/t of NO_x removed	Effect on Clinker Quality	Effect on Other Emissions
COM	High (all types of kilns)	15–30%	—	—	May be either positive or adverse	SO_x, CO, THC may increase
LNB Indirect-fired Direct-fired Kilns	High (all types of kilns)	15–30%	110–160 370–590	340–570 1,280–2,050	May vary with installation; type of fuel may be detrimental	May vary with installation, CO, THC may increase
SAC	High (precalciner, preheater and long kilns	20–50%	120–160 180–220	250–400 660–940	None	CO, THC may increase
SNCR	Medium (precalciner kilns)	40–70%	610–1,250	1,220–1,690	None	Potential for NH_3, PM_{10} emissions
SCR	Low (all types of kilns)	70–90%	3,150–10,050	4,840–7,500	None	Potential for NH_3, PM_{10} emissions; SO_3 may increase

methods. There is evidence that waste-derived fuels, such as tires and solvents, also tend to decrease these emissions.

While the raw material and fuel input of SO_2 into a cement kiln is in the range of 5–12 kg/tonne, the limestone acts as a natural scrubber trapping 90–99 percent of the SO_2 in the clinker product. Emission rates are very site-specific, having even larger variability than NO_x emissions: 1–10 kg_{SO_2}/tonne in wet and long dry kilns and 0.5–2 kg_{SO_2}/tonne in preheater and precalciner units. NO_x and SO_2 emissions are often inversely related in the clinkering process.

Carbon dioxide emissions have also become an important issue, especially since the process of calcination liberates 500 kg_{CO_2}/tonne of clinker. The fuel combustion–related emissions add between 150 and 400 kg/tonne depending on kiln efficiency and fuel used. Total per unit CO_2 emissions will be reduced gradually over the next 20 years through plant efficiency improvements, the use of waste fuels otherwise landfilled, landfill gas as a secondary fuel, and the use of flyash in blended cements to reduce the need for clinker.

Units of emissions reporting vary widely, with the following being common:

- concentration ppmv, 10–11 percent oxygen
- mg per m^3
- kg_{NO_x}/tonne of clinker, or cement product

Each kiln has a characteristic exhaust gas flow rate to allow conversion to a mass-based criterion: older kilns, 2900–3700 m^3/tonne; preheater/precalciners, 2200–2500 m^3/tonne.

NO_x emission control technologies

Several methods of NO_x reduction were assessed, although only some of these have been proven commercially. These include both combustion technologies that reduce or prevent emissions at their point of generation and postcombustion methods that reduce emissions already generated. See Table C-10.

Combustion and operational modifications. Process optimization through combustion modifications (COM) was one of the first options to be considered for reducing emissions. Improvements to be considered include: improved process control and automation systems, reduced excess air and firing temperature, improved raw mix design that optimized clinker cooling and air preheat, and most other modifications that improve thermal efficiency. These low-cost options should be maximized but must be evaluated against degradation of clinker quality and possible increases in other emissions.

Low-NO_x burners. This is one of the main areas of interest in environmental performance of cement kilns. The use of low-NO_x burners (LNBs) is being studied internationally by research organizations and cement companies. Although the concepts of a staged burning zone with exit gas recirculation are commonly used for LNBs in power boilers, the higher temperatures required in kiln pyroprocessing make these methods more difficult to apply for an acceptable quality of clinker. Four types of low-NO_x burners (3 coal, 1 gas) were described in the Radian study, with a wide variety of results (5–30% NO_x reduction) on installed systems, depending upon:

- The initial baseline NO_x emission
- Whether excess air could be kept to design low values to maintain efficiency
- Length of testing time and resulting clinker quality
- Emissions of other pollutants

Operating experience and design will likely improve the concepts. It should be noted that while normal kiln coal burners receive their primary combustion air directly with the fuel, low-NO_x burners require an independent air supply (indirect firing) largely fed from the clinker cooler. On an existing kiln, this represents a significant cost compared to that of the new burner.

Staged air combustion. By breaking the heat input into two locations, short precalciner kilns avoid the need to transfer heat a great distance in a long kiln, thereby reducing emissions and fuel consumption. There is however some thermal and fuel NO_x formed by the fuel-rich precalciner burner. Studies are underway to mitigate this effect by firing some of the precalciner fuel in the initial kiln exhaust to create a reducing condition ($NO_x \rightarrow N_2$), with the remainder fired in the upper calcining section for complete burnout. Staged Air Combustion has the potential for about 20–30 percent emission reduction in new precalciner kilns, but these levels are more difficult to achieve on existing plants. Clinker quality in the upstream portion of the kiln would not be affected.

Selective noncatalytic reduction (SNCR). SNCR using injection of ammonia compounds is also being investigated in Europe as a feasible postcombustion technology. Solid urea or liquid ammonia injection into a high-temperature exhaust stream (900–1000°C) has been done on a trial basis, with a reported 30–60 percent NO_x reduction on precalciner kilns. Issues to be resolved include possible effects on ESP performance, sensitivity to injection location and gas temperature, increased N_2O, CO emissions and NH_3 slip, and the costs and effort in handling ammonia.

Other methods. A related technology, selective catalytic reduction, would operate with an ammonia injection upstream of a catalyst section in a lower temperature range of 300–400°C. This has not yet been employed on high dust and hostile exhaust stream applications such as cement kilns.

NO_x emissions can also be minimized through reductions in clinker/cement ratio, and tests with waste-derived fuels have in some cases shown a positive impact on reducing emissions. The overall environmental impacts of incorporating flyash or slag mixed into the finished cement product to reduce the need for clinker should be considered, as well as the use of substitute waste fuels such as solvents, tires, and landfill gases to supplement traditional fuels. A separate CCME publication was developed entitled "National Guidelines for the Use of Hazardous and Non-Hazardous Wastes as Supplementary Fuels in Cement Kilns." It deals mainly with a more stringent particulate level, as well as with hazardous air pollutants.

The conversion of older long wet or dry process kilns to larger capacity preheater and precalciner kilns will result in fewer emissions and lower fuel consumption per tonne of cement clinker. Note again that experience has shown that NO_x from coal-burning kilns is often lower than from gas-burning kilns, although other emissions may be higher.

General approach for a national emission guideline

The Working Group for the cement and lime industry consultation was established in Canada in March 1993. A total of 15 members were involved representing the federal and several provincial/regional governments, the two major industry sectors, and environmental groups.

The National Emission Guideline states that emissions from large new natural gas- or coal-fired cement kilns (capacity >1500 tonnes/day) should not exceed 2.3 kg of NO_x per tonne of clinker production, based on a monthly average time period. The monthly averaging period was chosen to avoid issues associated with the transient nature of the hourly/daily emissions profile of most kilns. Special permitting could be done for the use of other fuels such as oil or petroleum coke, or for the use of alternative fuel mixes including landfill gas.

For an existing large cement kiln, the New Source Guideline limits stated below shall apply when a modification results in a 25 percent increase in permitted kiln capacity. For modifications associated with one-time lesser increases in permitted capacity, a program to improve performance should take advantage of cost-effective technologies to achieve feasible emission reductions. The resulting emission levels on these modified kilns do not necessarily have to meet those limits stated for new kilns. For kilns smaller than 1500 tonnes/day, or where there are opportunities to improve environmental performance and energy efficiency on kilns for which major modifications are planned, emission rates and control methods can be evaluated in close consultation with the appropriate regulatory authorities.

The reduction of clinker production, or the offset of other emissions, is also recognized as a strategic option for the industry and for regulatory bodies to deal with permitting issues on existing or new plants. Where a plant proposes to reduce

clinker production by the addition of additives such as flyash or slag, a higher NO_x limit could be considered. This would be based on the portion of clinker produced from primary raw materials, excluding the additives. This allowance would serve to increase the kg/tonne of clinker emission level for the kiln, and to recognize a reduction in overall net emissions for a given amount of cement produced.

A similar approach was introduced to credit NO_x emissions if waste heat from the stack exhaust or clinker cooler was used for any other heating processes not normally associated with kiln operation.

Emissions of other pollutants. The cement industry has previously been mostly concerned with other emissions involving particulate matter, addressed in a 1974 regulation. It had been agreed to in the consultation process that a comprehensive evaluation of other emissions would be included to the extent that would be practical. When considering the installation of process modifications to reduce emissions of NO_x, other pollutants should be minimized to avoid adverse environmental impacts, while maintaining acceptable clinker quality. The guideline has a revised particulate limit of 0.2 kg/tonne of clinker (about 90 mg/m^3) from the stack, and 0.1 kg/tonne from the clinker cooling system. New kiln systems should be designed, and raw materials selected, to minimize SO_2 and CO emissions while remaining in compliance with NO_x guidelines.

Measurement. The need for cost-effective pollution prevention, efficiency improvements, and emissions measurement has created a need for new types of integrated control and measurement systems. One of the best methods to improve environmental performance initially is to maximize efficiency through optimization in their manufacturing and energy processes. Many cement kilns now use continuous NO_x measurement at the kiln exit to feed back information to the process control to optimize combustion. Capital costs are usually small, especially if PEM/optimization software systems can be included at the time of a control system upgrade. The increased operator awareness of the plant is a benefit, and fuel savings, quality control, and maintenance planning can quickly pay back initial costs.

New cement kilns should measure NO_x and SO_2 emissions using a continuous emissions monitoring (CEM) system. If the measurement system used for process control at the kiln exit is to be used for emissions reporting, the method should show these emissions to be representative of those exiting the stack. On existing kilns that are to undergo major modifications, measurement should be done with a CEM system, or by a method of comparable effectiveness to continuous monitoring. The averaging period, the reporting requirements, and the type of emissions monitoring are at the discretion of the local permitting authorities.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.

Centrifuges

Centrifuges are used in applications as diverse as separating sand from liquid (a mixture of oil, water, and caustic soda) in processing oil sands to milk creaming in agriculture.

Centrifugal separators are used in a wide variety of applications, some of them conventional, such as food processing, for instance dairy creamers. More recently, the design demands placed on this technology have increased with mineral

beneficiation applications, such as extracting synthetic crude from oil sands. When the prototypes of this application were first tried in 1977, the centrifuge nozzles had a life of a few hours. The material had to be changed to one that better withstood the highly abrasive sand involved. Nozzle angle also had to be varied.

Basic Working*

Frequently, mixtures of solids and liquids must be separated into their components in order to be effectively utilized. The mixtures may be of different solids or the liquid fraction may contain dissolved solids that are to be removed. Such situations occur in food processing, mineral beneficiation, and chemical conversions.

When the solids and the density difference is small and the flow volume is large, disk nozzle centrifuges are often the best means to accomplish the purification. The separation, which takes place within the rotor of a disk nozzle centrifuge, is effected by the G force, the "rising rate" of the liquid (related to the feed flow) and the separation area provided by a set of conical, close-spaced disks as well as the process factors of fluid viscosity, particle size, shape, and density. In addition, the design of the equipment must allow for the quantity of solids to be handled, the flow characteristics of the slurry, and other practical engineering considerations.

Whereas disk nozzle centrifuges have been in use for concentration purposes for a long time, they are now employed as purifiers; in which instance a large flow of "upflowing" liquid greatly enhances the purity of the products.

Using the elutriating stream concept, we can deduce the beneficial action of displacement washing versus dilution washing. The improved flow pattern enhances the "classification" of particles. Three examples are presented that show typical processes.

Disk nozzle centrifuges have been used for over 60 years for the concentration of fine solids in a stream of slurry feed. Such centrifuges are now in common use around the world for handling food products, chemicals, minerals, biological materials, and waxes. These centrifuges are made in a variety of materials and sizes and in many different countries by various manufacturers with differing design concepts. However, the significant principles are well established.

In addition to simple sedimentation, where the objectives are to obtain a clarified effluent or thickened solids-loaded fraction or the separation of two liquid phases, it is also possible to simultaneously introduce a stream of "wash" into the centrifuge. There may be several purposes served. First, the discharging solids may exit in the "wash" fluid rather than in the mother liquid, or upflow action of the wash stream may flush out a smaller size solids fraction from the larger size solids fraction. In the first case we have purification by washing (solubles removal) and in the second case we have purification by classification (slower-settling solids removal).

Figure C-19 is a photo of an intermediate size disk nozzle centrifuge. This machine is fitted to operate under elevated temperature and pressure conditions and for the purification of terephthalic acid crystals. Note the electric motor, the overhead V-belts, and the flexibly mounted bearing assembly. These power the pendulum-suspended rotor, which has somewhat of a double cone shape with the nozzles located at the largest periphery. The housing material is Hastelloy for extra corrosion resistance and the rotor is similarly special to withstand the severe mechanical and chemical conditions.

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* Source: Dorr-Oliver Inc., USA. Adapted with permission.

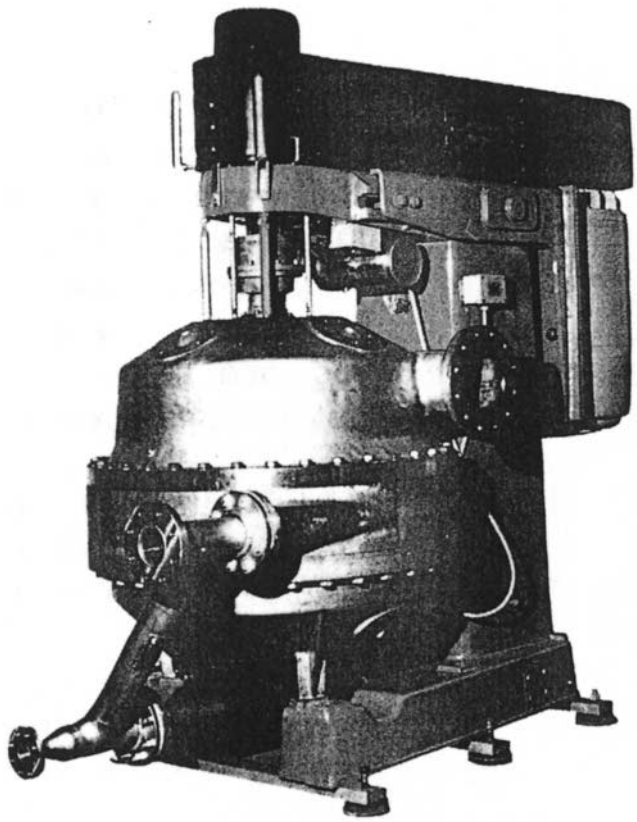


FIG. C-19 Photo of Merco® PCH-30 centrifuge. (Source: Dorr-Oliver Inc.)

Figure C-20 is a cutaway view of a disk nozzle centrifuge that shows the flow pattern. It is easy to follow the path of the feed slurry as it flows continuously down into a central rotating feed distributor and laterally into the main separating chamber. Here the high sedimenting force (of perhaps 5000 g) acts to draw the heavier solids outward where they discharge from the rotor through backwardly reacting nozzles. This slurry is then gathered in a collecting volute and recycles (by means of its velocity head) back to a reinjection port in the bottom of the stationary housing and jets back into the rotor hub where it is reaccelerated.

A major portion of the underflow can be drawn off through a valve located appropriately in the return loop. Meanwhile, the surplus flow (the feed minus the draw-off) moves inwardly through the separating disks, where fine solids are removed, and it discharges from the top of the rotor as clarified overflow. The method of feeding into the disk stack through a set of vertically punched holes and the arrangement of spaces on the disks are well known. They are sized and located in a specific fashion appropriate to the application. Similarly the recycled flow is directed through special tubes back toward the nozzle region.

Figure C-21 is a cutaway view of the centrifuge with a special wash inlet system inserted at the bottom of the housing. This system makes it possible to inject large volumes of wash at the interior of the rotor where the flow must travel inwardly and countercurrently to the outward motion of the solids. This action can accomplish a great increase in the washing capability of a single stage or it can significantly enhance the sharpness of the separation between two classes of solids.

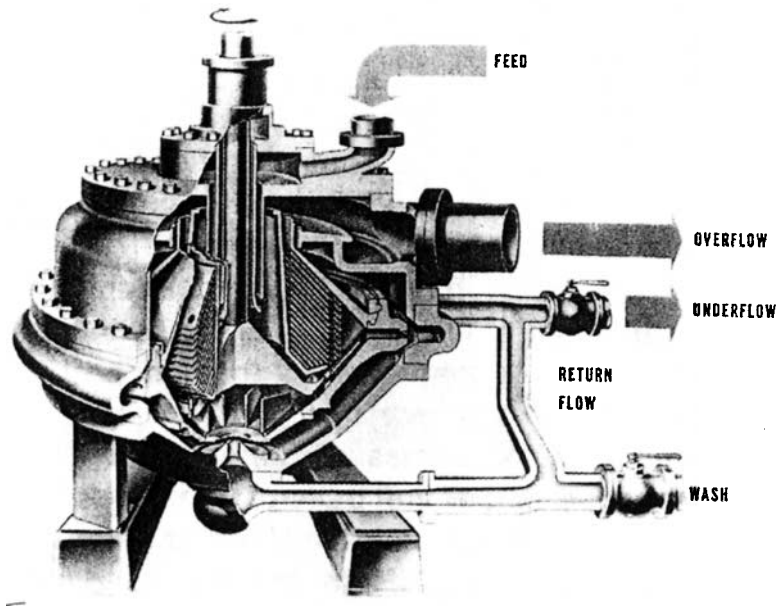


FIG. C-20 Cutaway view of a disk nozzle centrifuge and its flows. (Source: Dorr-Oliver Inc.)

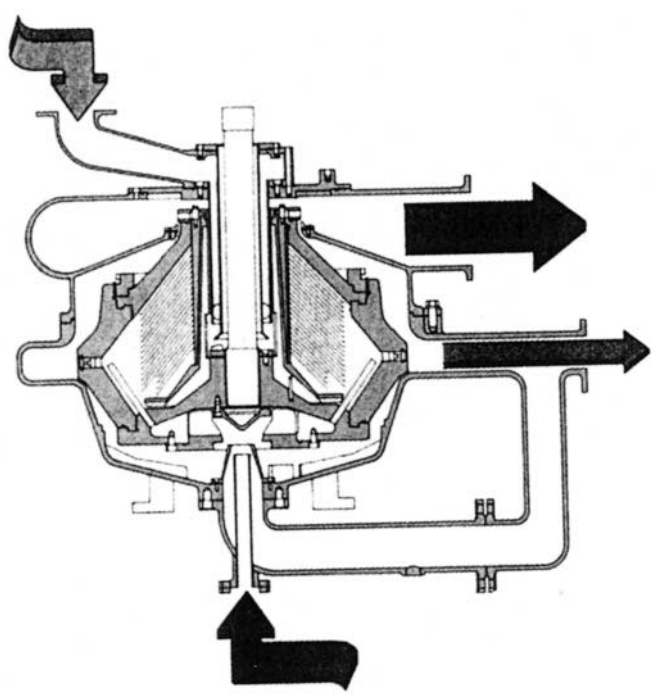


FIG. C-21 Cutaway view of Merco® centrifuge with a special wash inlet. (Source: Dorr-Oliver Inc.)

In the process of washing, the most important thing is to remove the contaminants as completely as possible. Thus, the use of large quantities of wash are generally of benefit. However, we often want to conserve the wash fluid for economic reasons. Accordingly, a balance is struck and the degree of efficiency of washing becomes important.

In the process of classification, the most important thing is to remove all of the slow settling solids but to *not* remove the other solids. Thus, the appropriate quantity of wash has to be sought by testing.

Reference and Additional Reading

1. Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

Ceramics

Ceramic is a porcelainlike material that has better corrosion resistance than most metals, so ceramics are used to coat items such as turbine blades. Ceramics used to be prone to cracks due to brittleness, since ceramics lack the ductility of metals. Product improvements in this material have made major strides recently, and operating temperatures for ceramics keep rising. Various manufacturers have their own customized ceramics and their manuals and specifications should be consulted if there is a question or an overtemperature problem suspected.

Chemical Cleaning (*see also* Cleaning; *cleaning information in many other sections*)

The range of chemical cleaners (routine and also used preparatory to most overhaul and repair processes) has grown recently to bewildering proportions. Of prime consideration on the overhaul facility's requirement list is lack of carcinogen content and other increasingly enforced environmental regulations. A facility setting up cleaning facilities should consider not only existing legislation but also potential or proposed changes in environmental law before investment.

The end user's requirement for time between overhauls is greatly reduced with an effective wash system for his or her engine. If the system can be used online, the advantage is still further enhanced. Wash systems are generally designed and manufactured by firms that specialize in this work. Each turbine model's reaction to a system is different. Nozzle size and angle, fluid rate, and so forth have to vary with each model for maximum effectiveness. A manufacturer with highly successful wash systems might find that they fail entirely with a new engine model. Most OEMs generally put their own nameplates of "ownership" on their subsupplier's wash systems.

Chemical Complex; (Petro)Chemical Complex; Chemical Plant

A *petrochemical complex* (a specific kind of chemical complex) is generally a large facility that may encompass more than one company. The complex may process upstream petroleum products (primarily oil and natural gas) into complex downstream chemical and plastic products. As an illustration, a product flowchart (Fig. C-22) from the Petrochemical Company of Singapore (PCS), indicating numerous downstream companies, is included. The large number of companies and products resulting from the complex should be noted. (All acronyms used

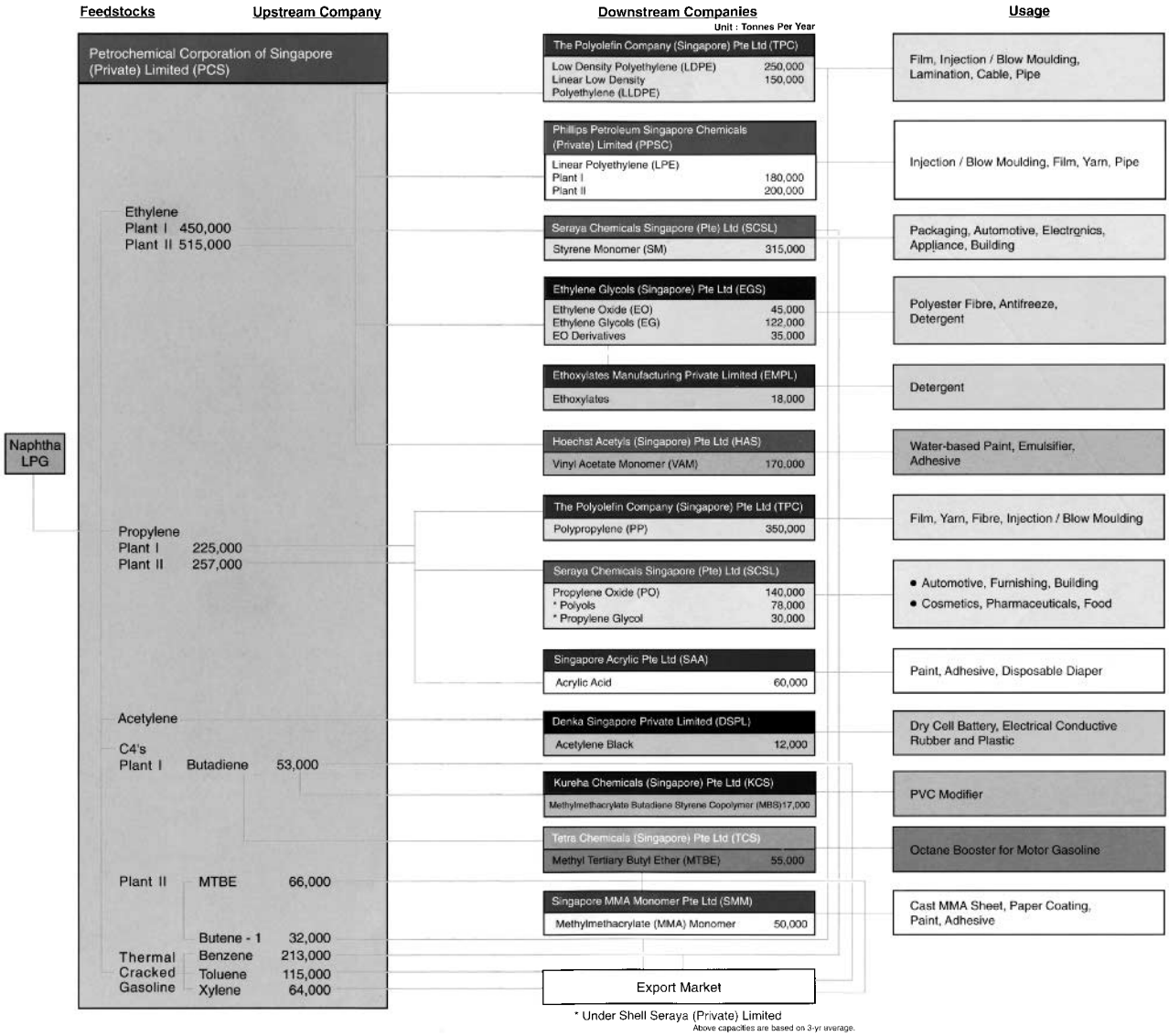


FIG. C-22 Production schematic of overall PCS complex. (Source: Petrochemical Company of Singapore.)

are described in full on the flowchart.) Note that PCS was commissioned in two phases, phase I and phase II. Most chemical complexes evolve or grow in this way, with the simpler products produced initially and more sophisticated ones in later phases. Or the later phase may be constructed for larger volumes of the initial products. A case study at this point is presented to provide insight. PCS phase I and II are described as an example. Figure C-23 is a summary of phase I of the PCS plant itself. Figure C-24 indicates the physical layout of all the various plants within the complex in both phases. Figure C-25 illustrates the ethylene plant in PCS's phase II.

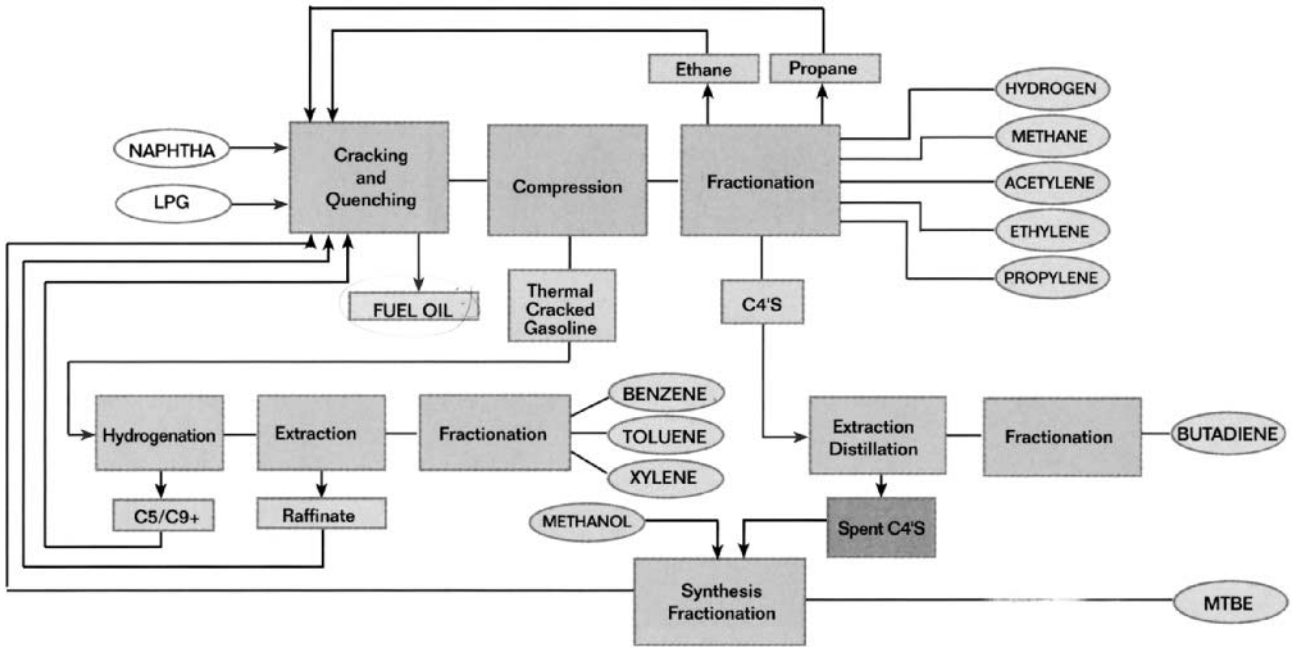


FIG. C-23 Schematic of process flow (PCS Phase I). (Source: Petrochemical Company of Singapore.)

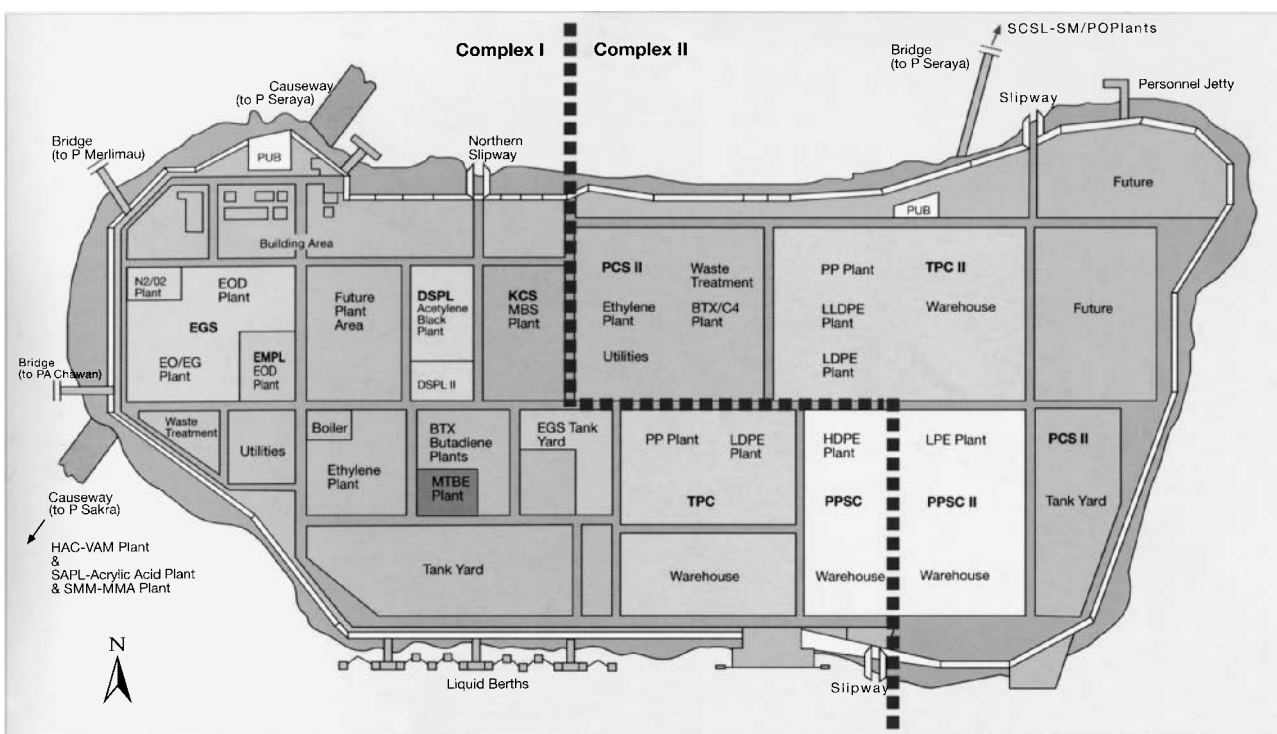


FIG. C-24 Layout of Complex I and II. (Source: Petrochemical Company of Singapore.)

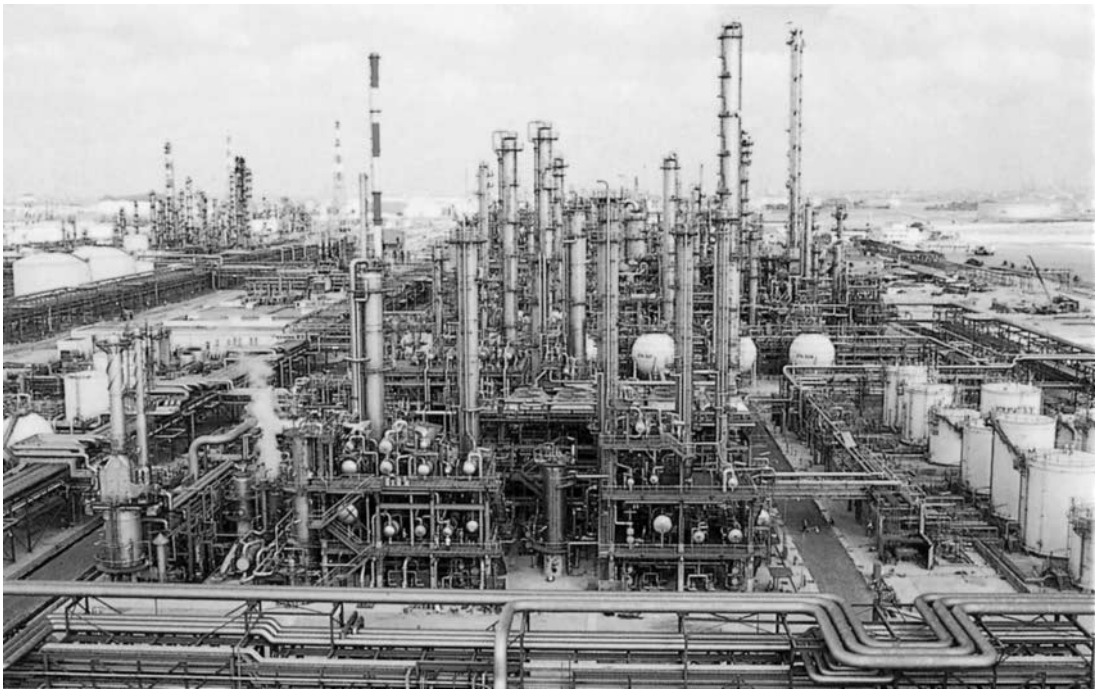


FIG. C-25 PCS II ethylene plant. (Source: Petrochemical Company of Singapore.)

"Complex II" Project Summary

The expansion of the Singapore Petrochemical Complex, known as the Complex II project, was launched on March 1, 1994. This S$3.4 billion project involves:

- The expansion at Pulau Ayer Merbau of the existing plants of PCS, TPC, PPSC, DSPL, EGS, and KCS.
- The establishment of a new downstream Styrene Monomer/Propylene Oxide (SMPO) plant by Seraya Chemicals Singapore (Private) Limited (SCSL) on nearby Pulau Seraya.

With the on-schedule completion and startup of the expansion project in April 1997, PCS's ethylene capacity is now doubled to close to 1 million tonnes/year and it is one of the largest leading ethylene producers in the region.

In the meantime, three more new downstream projects sited on the adjacent Pulau Sakra were added to the expansion. They are:

- A Vinyl Acetate Monomer (VAM) plant by Hoechst Acetyls Singapore Pte Ltd (HAS), which started up in August 1997.
- Sumitomo Chemical's joint venture (with Sumitomo Seika and Toagosei) plants to produce acrylic acid (and its derivatives) as Singapore Acrylic Pte Ltd (SAA).
- Singapore Monomer Pte Ltd (SMM), another Sumitomo Chemical joint venture (with Nippon Shokubai) producing methylmethacrylate (MMA) monomer.

Chemicals

Chemicals can be either organic or inorganic. The range of chemicals and chemical compounds in use today, even if listed by name only, would fill several volumes.

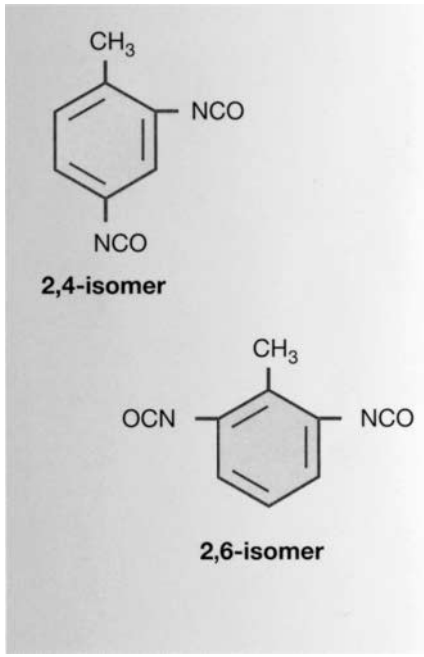


FIG. C-26 Structural formulas of TDI isomers. (Source: ARCO Chemical.)

For illustrative purposes in this handbook, a few specific illustrative examples of applications common to process engineering and some associated technology will be discussed. Toxic chemical handling technology is one such example.

Chemicals (Toxic), Handling

The illustrative example used in this case is an organic chemical that results from hydrocarbon processing. In actual fact, every toxic chemical, inorganic or organic, requires specific precautions. The example that follows indicates the extent of the complexity and caution required with handling a toxic chemical. For liability reasons, manufacturer-provided procedures should be strictly followed.

Some chemicals resulting from hydrocarbon petroleum product processing are toxic and require extreme care during all handling processes. For illustrative purposes, information on handling ARCO Chemical's toluene diisocyanate follows.*

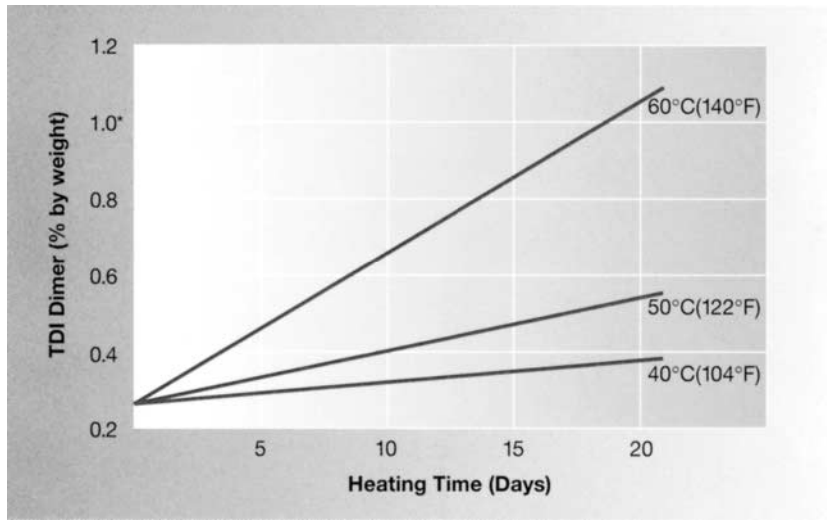
Example: Toluene Diisocyanate (TDI)

ARCO Chemical's toluene diisocyanate is referred to as TDI-80 because it is an 80:20 mixture of the 2,4- and 2,6-isomers of TDI. Structural formulas of these isomers are shown in Fig. C-26. Also see Fig. C-27.

ARCO Chemical produces TDI-80 in two forms, designated Type I and Type II. Both have the 80:20 isomer ratio, but they differ slightly in acidity and hydrolyzable chloride content.

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* Source: ARCO Chemical, USA. Adapted with permission.



* Dimer crystals precipitate at ambient temperature when dimer concentration approaches 1%.

FIG. C-27 TDI dimer formation over time at various temperatures. (Source: ARCO Chemical.)

Type I is used in foam and nonfoam urethanes. Type II is used in nonfoam urethanes, rebonded flexible foam, and other applications.

Physical properties of TDI-80, Types I and II, are shown in Fig. C-28. Those properties marked by an [a] are ARCO Chemical specifications; other properties are those typical of commercially available TDI.

TDI has a sharp, pungent, sweetish odor. Its vapors are toxic. Certain precautions are necessary when handling or using toluene diisocyanate. Before using TDI, obtain and study ARCO Chemical's *Material Safety Data Sheet* (MSDS) and product literature. For more information, see "TDI Safety and Handling" below.

Reactivity

ARCO Chemical TDI is a clear liquid, water-white to light yellow in color. It yellows on exposure to light.

Chemical: TDI reacts readily with compounds containing active hydrogens, such as acids and alcohols. Contact with bases, such as caustic soda or tertiary amines, might cause uncontrollable polymerization and rapid evolution of heat.

Water: On contact with water, aromatic polysubstituted ureas are formed, and carbon dioxide plus heat are evolved. In time, white aromatic polyurea crystals will precipitate.

Heat: High temperatures can cause formation of dimer and discoloration of the TDI. This phenomenon is time and temperature-related (see Fig. C-27).

When the level of dimer approaches 1 percent by weight, solid dimer forms as needle-like crystals. These crystals cannot be completely filtered out because the solution is supersaturated and new crystals are formed to replace those that are removed.

Temperatures below 15°C (59°F) cause TDI to freeze. Frozen TDI is also white and crystalline. If frozen, TDI may be thawed by heating (see "Thawing TDI" below for methods and proper precautions).

Note: As can be seen from the above discussion, if white crystals are detected in TDI, they may be frozen TDI, aromatic polyurea, or dimer. For suggestions on

Molecular Weight	174.163
Assay[a], min (%)	99.7
Isomer Ratio[a] (%)	
2,4-isomer	80±1
2,6-isomer	20±1
Acidity[a], as HCl (%)	
Type I	0.002-0.004
Type II	0.008-0.010
Hydrolyzable Chlorides[a] (%)	
Type I	0.003-0.008
Type II	0.011-0.014
Chlorine[a], max (%)	0.20
Ash (ppm)	20
Color (APHA)	15-40
Specific Gravity @ 25/25°C [77/77°F]	1.22±0.01
Density (lbs per gal)	
@15.5°C [60°F]	10.23
@20°C [68°F]	10.14
@38°C [100°F]	10.02
@60°C [140°F]	9.86
Viscosity (cs)	
@50°C [122°F]	1.5
@100°C [212°F]	0.8
@135°C [275°F]	0.5
Melting Point Range	
(°C)	11.5-13.5
(°F)	52.7-56.3
Freezing Point	
2,4-isomer (°C)/(°F)	15.0/59.0
2,6-isomer (°C)/(°F)	7.2/45.0
Boiling Point	
@10mm Hg (°C)/(°F)	121/250
@760mm Hg (°C)/(°F)	251[b]/484[b]
Flash Point[c], COC (°C)/(°F)	132/270
Fire Point, COC (°C)/(°F)	142/288
Latent Heat of Evaporation (Btu/lb)	
@120°C [248°F]	131
@180°C [356°F]	121
Vapor Density, air = 1	6
Vapor Pressure, approx. (mm Hg)	
@20°C [68°F]	0.01
@120°C [248°F]	11
@130°C [266°F]	16

[a] ARCO Chemical specification

[b] Decomposes

[c] The flammability properties of this material (or any other material) are not intended to reflect the fire hazards presented by any resultant cellular or foamed plastic product.

FIG. C-28 Physical properties of TDI-80 produced at the Lake Charles, La., plant. (Source: ARCO Chemical.)



FIG. C-29 A drum for TDI containment. (Source: ARCO Chemical.)

dealing with such situations, see "What to Do In Case of . . ." below or call the manufacturer.

TDI Shipments

ARCO Chemical TDI may be obtained in tank cars, tank trucks, cylinders, or drums from this plant or various worldwide distribution centers and terminals. For export, ARCO Chemical has the capacity to ship TDI in bulk and full container lots of drums via ocean vessels.

Cylinders: In the United States, ARCO Chemical provides TDI in carbon steel cylinders that contain 230 U.S. gal and are used at 20–30 psig. Some cylinders with a capacity of 263 U.S. gal and a pressure rating of 275 psig are also available. Intended to be moved with a forklift, the cylinders have two-way-entry metal skids.

Tank Cars: TDI is most frequently shipped in 20,000-gal cars, although other sizes are available upon request. All cars are insulated and have exterior heating coils. All cars are padded with nitrogen.

Specific arrival temperatures with tank car deliveries cannot be guaranteed.

Tank Trucks: TDI is shipped in 4000- to 5000-gal trucks. Shipment weights range from 40,000 to 50,000 lb, depending on the point of origin and road weight regulations.

TDI trucks are equipped for top-unloading only and have compressors and air driers to maintain product integrity. Pumps can be made available upon request. Tanks are constructed of stainless steel; all are insulated and have exterior heating coils.

Drums: TDI is available in 55-gal nonreturnable drums, made of 18-gauge steel (minimum), with phosphatized interiors. Drums contain 551 lb (250 kg) of TDI (see Fig. C-29).

Ocean Vessels: Large chemical companies have the capability to serve world markets with shipments of large quantities in bulk or in drums.

Unloading TDI

Toluene diisocyanate is regulated by the Department of Transportation (DOT) as a Packaging Group II Toxic. Since TDI can cause serious injury to the lungs, eyes, and skin, all persons near the unloading site must wear protective clothing and equipment. They must observe the safe handling procedures and practices



FIG. C-30 A TDI tank car. (Source: ARCO Chemical.)

prescribed in ARCO Chemical's *Material Safety Data Sheet* (MSDS) and product literature. "TDI Safety and Handling" below should be carefully read by, and explained to, all employees. For additional employee training, large chemical companies offer videotapes covering handling procedures.

Customers should give careful consideration to the way that TDI will be received. Adequate facilities must be provided (see "Storage of TDI" below). Ample water should be available at the unloading site, including a shower equipped with a quick-opening deluge head and an eyewash fountain.

The site should also be equipped with an inert gas such as nitrogen or dry air for use in padding the car and purging lines.

Note: While nitrogen is preferred, all future references to "inert gas" should be taken to mean either nitrogen or dry air (–40°C/°F dew point), and all references to nitrogen should be taken to mean that dry air may also be used.

Unloading tank cars

TDI tank cars. Specific manufacturers generally operate a large fleet of dedicated TDI tank cars (see Fig. C-30). Both general-purpose and modified DOT IIIA cars are currently in service. The modified DOT IIIA cars have the following features:

- $^{9}/_{16}$ in tank shell thickness
- Full $^{1}/_{2}$ in protective headshields
- No bottom valve outlet; top unloading only
- All stainless steel fittings
- Safety valves; not safety vents
- All top fittings are mounted on a 20-bolt cover plate, inside a protective housing (bonnet)
- Hot-dipped galvanized steel safety grating
- Two dip legs (eduction pipes), either can be used for unloading (see Fig. C-31)

Most tank cars have a capacity of 20,000 gallons. Figure C-32 shows a typical arrangement of the fittings found under the bonnet on the top of the tank car. In

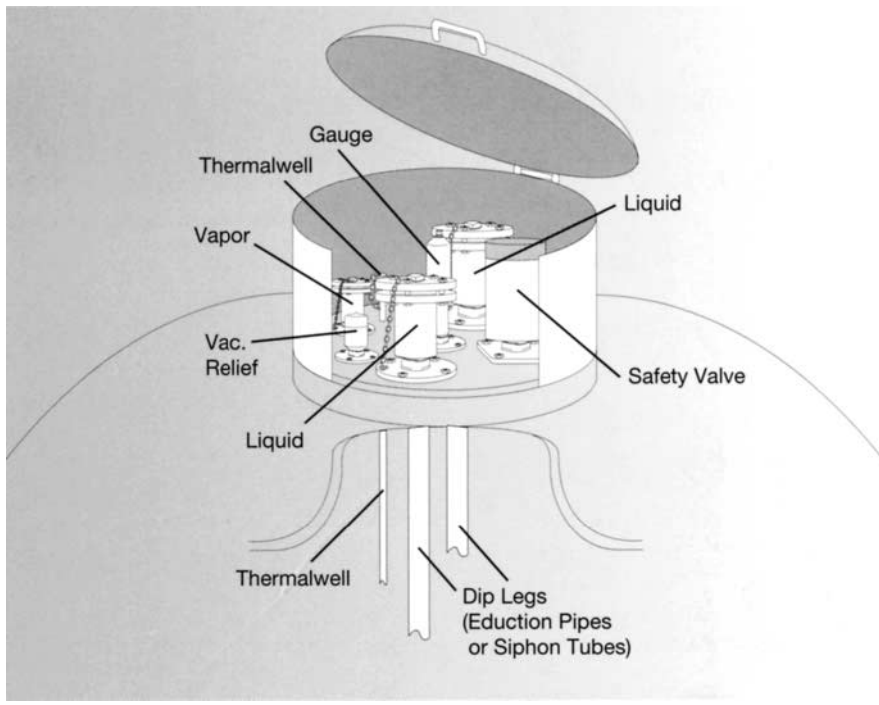


FIG. C-31 Tank car dip legs. (Source: ARCO Chemical.)

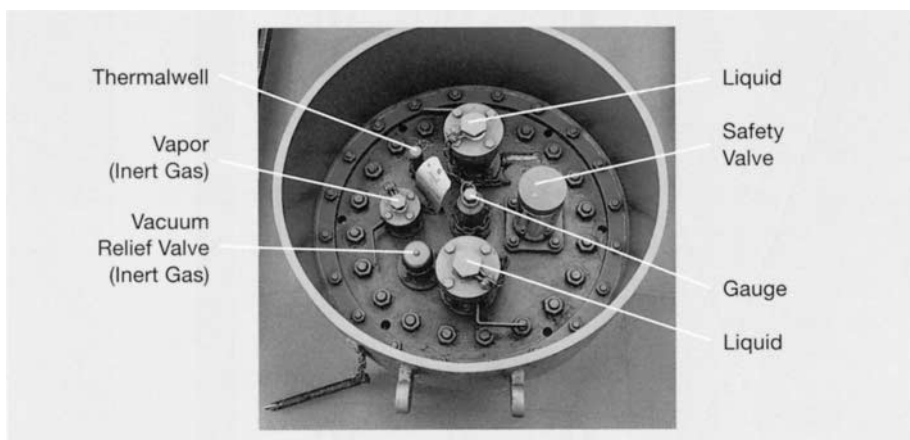


FIG. C-32 Top unloading connections (for modified DOT IIIA cars). (Source: ARCO Chemical.)

addition to these bonnet fittings, every car, regardless of type, has a manway and safety valve. Every car has a thermalwell under the bonnet, which is used in taking the temperature of the car's contents.

All chemical cars made by this information source are designed for top unloading through either eduction pipe. (See Fig. C-32 for typical connections.)

TDI cars are insulated to prevent freezing. However, in the event freezing does occur, all cars have external steam coils for thawing the TDI (see "Thawing TDI" below).

Preliminary procedures. Before tank cars or tank trucks are unloaded, all workers must put on proper protective clothing and equipment. The following three steps should then be taken.

1. *Depressurize the Car:* Open the ball valve on the 1-in inert gas inlet located on top of the car (see Fig. C-32).
2. *Take TDI Temperature:* Temperature is taken through a thermalwell, which is located between the 1-in inert gas inlet and the 2-in eduction pipe. Insert a thermocouple into the thermalwell and read the temperature. (The use of a conventional thermometer may result in an erroneous reading because the ambient temperature is usually lower than the internal TDI temperature.) *If the gas inlet valve is used for taking the temperature, a self-contained breathing apparatus must be worn as protection from TDI vapors.*

   TDI-80 is normally loaded into insulated tank cars or tank trucks at 24–30°C (75–86°F); in winter it is loaded at 38–43°C (100–110°F). Recommended unloading temperature is 21–30°C (70–86°F). If the temperature is between 17°C and 21°C (63–70°F), the TDI can be heated. If the temperature is below 17°C, it is likely that there is some freezing, and the TDI must be thawed.
3. *Sample the Car Contents:* After the car or truck has been depressurized and the TDI temperature measured, a sample should be taken for testing. While this is being done, full protective clothing and a self-contained breathing apparatus must be worn.

   For tank cars, the preferred procedure is to take a sample from the unloading line (through a customer-installed value). This avoids opening the manway cover and losing the nitrogen pad, and thus eliminates a possible source of contamination.

   If a sample is taken through this valve, first flush out 1–5 gal of TDI (for proper disposal procedure, see "Handling Spills and Leaks" below). Flushing ensures that a representative sample is being taken. This is particularly important in determining if aromatic polyurea or dimer (white precipitate) is present.

   If a sample must be taken directly from a pressurized car or truck manway, be sure it is an "all-level" sample at or near atmospheric pressure. Car hatches should be open for as little time as possible. During inclement weather, make provisions to prevent contamination of the product.

   An all-level sample is taken by using a clean, dry, amber-colored glass bottle in a weighted bottle holder. Be certain that workers are wearing proper protective gear before and during sampling.

   To be sure of getting a representative sample, the bottle holder should be lowered to the bottom and then withdrawn at such a rate that the bottle is not quite full when it reaches the surface. (This may take some practice.) Keep the sample out of direct sunlight to prevent yellowing.

   The filled sample bottle should be capped, cleaned, and plainly labeled with product lot numbers, tank car or truck number, compartment number (if more than one), date, and sampler's initials.

What to do in case of . . . *White Precipitates:* There are three causes of white precipitates in TDI: dimer (caused by excessive heat), aromatic polyurea (caused by the presence of water), or frozen TDI. If it is not obvious which of the three is present, heat the crystals. If they melt at 16–21°C (60–70°F), they are frozen TDI. If they melt at 150–160°C (302–320°F), they are dimer. If they do not melt, they are aromatic polyurea.

If the crystals are frozen TDI, the product can be thawed, remixed, and used. If the crystals are aromatic polyurea, they can be filtered out and the remainder of the TDI can be used. However, if the crystals are dimer, they cannot be completely removed (dimer reforms on filtration). The TDI should not be used because the dimer will affect urethane physical properties. It will clog lines and foam heads as well. If dimer is present, contact the manufacturer.

Discoloration: Normal TDI is water-white to pale yellow in color. A darker color means the TDI has been exposed to light or high temperature. A color something other than white to yellow means the TDI has been contaminated and should not be used. Call the manufacturer for assistance.

If the color has merely darkened, assume the cause is high temperature. (The chances of light-induced discoloration are negligible.) Since the high temperature may also cause dimer formation, the TDI should be tested. Simply cool a sample to room temperature. If white crystals precipitate, dimer is present and the TDI should not be used. If no white crystals are present, the TDI may be used. The discoloration will not affect physical properties or foam color.

General unloading regulations and suggestions. Department of Transportation regulations for unloading tank cars are given in Section 174.67 of Title 49, *Code of Federal Regulations, Hazardous Materials Regulations*. The regulations require that all persons responsible for tank car unloading should be familiar with these regulations and that all applicable requirements should be observed.

Below are some of the pertinent federal requirements. Following several of them are related suggestions and recommendations, which this information source believes are also necessary or important to follow, even though they may not be part of the regulations. These are printed in italic type. The most important recommendation that this chemical manufacturer makes is that workers be familiar with the health and safety aspects of TDI, and that they use the proper protective equipment when contact with this product is possible.

1. Unloading operations must be performed only by reliable persons properly instructed in unloading hazardous materials and made responsible for careful compliance with this part.
2. Brakes must be set and wheels blocked on all cars being unloaded. *Tank cars should also be protected during unloading by such means as derails or locked switches*.
3. Caution signs must be so placed on the track or cars to give necessary warning to persons approaching the cars from the open end of a siding. Signs must be left up until after the cars are unloaded and disconnected from the discharge connection.

   The signs must be of metal or other comparable material, at least 12 in high by 15 in wide, and must bear the words, "STOP—Tank Car Connected" or "STOP—Men at Work." The letters are to be white on a blue background, with the word "STOP" at least 4 in high and the others at least 2 in high.

   If the unloading area has heavy traffic, it should be roped off and passersby warned by posting "Danger—TDI" signs. The contents of tank cars should only be unloaded during daylight hours or when adequate lighting is provided.
4. Unloading connections must be securely attached to unloading pipes on the dome outlet . . . before any discharge valves are opened.

   Tank cars must be depressurized before making any unloading connections.
5. Tank cars may not be allowed to stand with unloading connections attached after unloading is completed. Throughout the entire period of unloading,

and while the car is connected to the unloading device, the car must be attended.

6. If necessary to discontinue unloading a tank car for any reason, all unloading connections must be disconnected. All valves must first be tightly closed, and the closures of all other openings securely applied.

   Before disconnecting—for any reason—all lines should be cleared of liquid material by blowing with nitrogen or dry air.

7. As soon as a tank car is completely unloaded, all valves must be made tight, the unloading connections must be removed, and all other closures made tight, except for heater coil inlet and outlet pipes, which must be left open for drainage. If it has been opened, the manway cover must be reapplied by the use of a bar or wrench, the outlet reducer and outlet valve cap replaced by the use of a wrench having a handle at least 36 in long, and outlet valve cap plug, end plug, and all other closures of openings and their protective housings must be closed by the use of a suitable tool.

TDI tank cars can only be unloaded from the top, through either of the 2-in eduction pipes (dip legs).

Figure C-33 shows how unloading is accomplished using nitrogen. This dry-atmosphere padding is necessary to prevent a reaction between the TDI and any water vapor that might be present. Under no circumstances should a combustible gas be used; it presents an explosion hazard.

All fittings should be inspected for evidence of actual or potential leaks before the tank and piping system are pressurized. An oil trap should be installed on the inert gas supply line.

Tank cars are protected by a safety valve, set to relieve at 35 or 75 psig (depending on the car). The pressure system should be designed so as not to exceed a safe working limit; a pressure of 10–20 psig is recommended.

The steps involved in positioning the car and installing the necessary safety devices must be carried out in accordance with the regulations set forth in Section 174.67, as outlined previously. Before starting to unload, follow the instructions for depressurizing the car, taking the temperature, and sampling, under Preliminary Procedures, also previously. Then:

1. Be sure the tank car manway is secured. Make sure the storage tank is adequately vented.
2. While the 1-in inlet valve is closed, remove the plug and connect the inert gas line. (See Figs. C-32 and C-33.)
3. The temperature of the unloading line should be 21–30°C (70–86°F), the proper temperature for unloading. Check the line temperature and preheat the line, if necessary, before connecting it to the 2-in unloading valve, which leads to the eduction pipe.
4. Open all valves in the unloading line.
5. Open the inert gas supply valve. The pressure on the car will be effectively established by the setting of the inert gas pressure-regulating valve. The flow of TDI can be controlled by a valve in the unloading line.

After unloading is complete (or if unloading must be interrupted):

1. Purge the unloading line with nitrogen before disconnecting. Equalize the line pressure. Close all valves in the line. Disconnect the unloading valve and cap it.

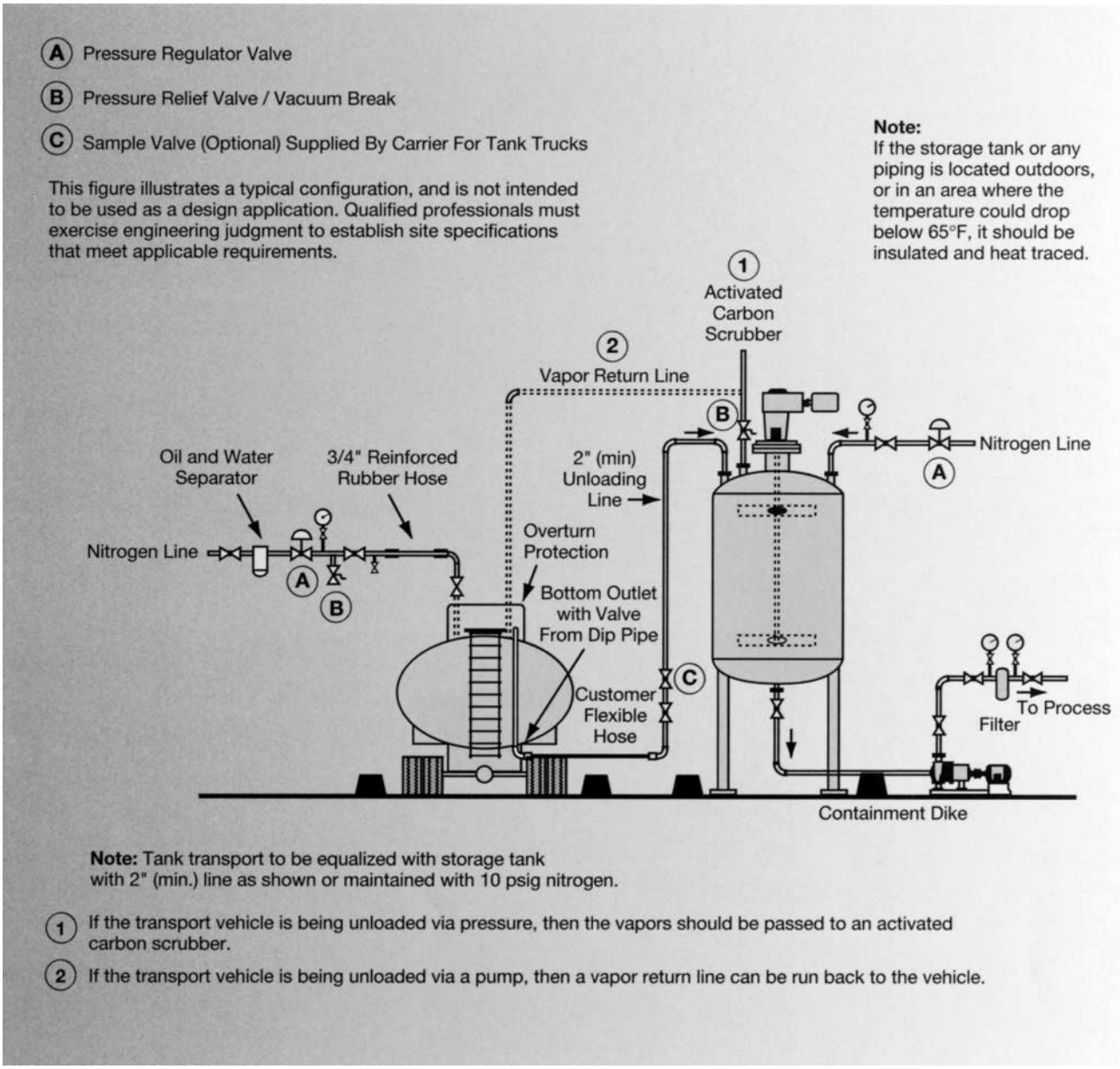


FIG. C-33 Top unloading and storage arrangement. (Source: ARCO Chemical.)

2. Disconnect the steam lines and purge the coil by blowing with nitrogen. Do not replace the caps on the steam line.
3. Repressurize the car with nitrogen to 5–10 psig.
4. Secure the dome bonnet.
5. Be sure all four placards are in place before returning the car by the prescribed routing.

Unloading tank trucks. Prior to unloading, it is the recipient's responsibility to provide competent and knowledgeable supervision, safety equipment, and a properly designed unloading area. Tank trucks are unloaded by the driver of the



FIG. C-34 Tank truck unloading. (Source: ARCO Chemical.)

vehicle, who is responsible for following the proper safety rules, as prescribed by recipient, by the manufacturer, and by government regulations. Trucks are specially equipped for unloading as shown in Figs. C-34 and C-35.

The unloading area must be large enough for easy turning and positioning of the vehicle. It should be level, to ensure complete unloading. It must be covered with an impervious material, such as concrete or steel plate (not asphalt) to prevent ground contamination in the event of a spill. The area also must be contained to prevent a spill from spreading. Safety showers and eyewash stations must be nearby.

The supervisor should make sure the unloading area is clear and that adequate facilities are ready for receiving the shipment. Before unloading begins, the supervisor must check the temperature of the TDI (and adjust it, if necessary). When the temperature is within the proper limits, it is recommended that the supervisor take a sample of the shipment.

After unloading is complete, all lines should be purged with nitrogen. The tank truck should then be padded with nitrogen (3–5 psig).

Unloading TDI cylinders

The cylinders are equipped with the following:

- Primary liquid dip tube fitted with a $1^1/_4$-in Stratoflex fitting liquid dip tube fitted with a $^1/_2$-in Stratoflex fitting
- $^3/_8$-in vent valve with a bleed-down cap assembly
- $^3/_8$-in nitrogen valve with a snaptight fitting and check valve
- Level gauge reading 5–95 percent volume
- 150 psig pressure safety valve

Note: Fitting types and sizes may vary.

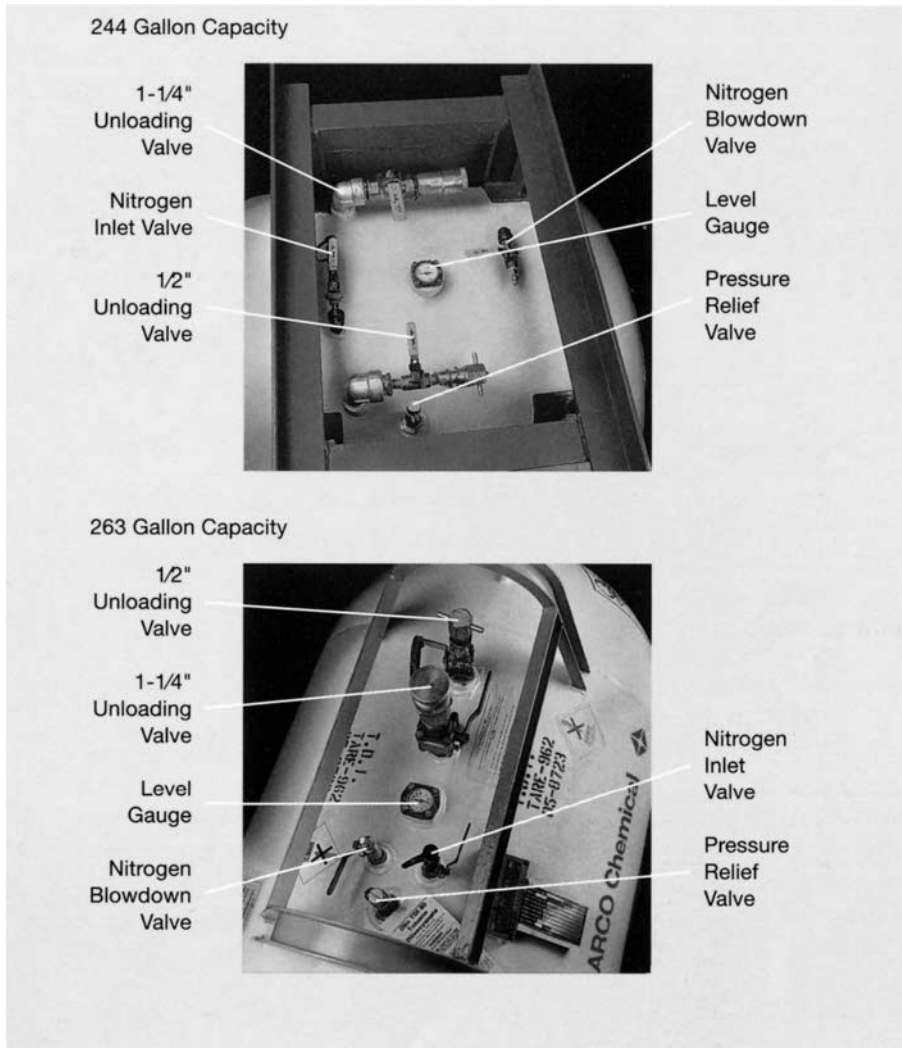


FIG. C-35 Top of cylinders. (Source: ARCO Chemical.)

Receiving cylinders. *Leakage/Damage:* Cylinder exteriors are cleaned and inspected prior to shipping so that damage can be readily seen. Upon receipt of a cylinder, check for any external damage or leakage. As long as there is no leakage, the cylinder can be accepted. Make a note on the carrier's bills and send a copy of the bill of lading and damage report to the manufacturer. If the cylinder is leaking, call the manufacturer and follow the steps in its emergency response guide. Report any dents or damage to skids or cowling to the manufacturer.

Pressure: Cylinders should have a positive nitrogen pad pressure in the range of 5–25 psig. If no pressure is present, call the manufacturer for instructions.

Returning cylinders. *Preparing Empty Cylinders for Return:* Be sure that the dust caps are tightly screwed onto the male and female self-sealing couplings and the nitrogen inlet caps are in place when the tanks are not in use. This is essential to prevent possible contamination and vapor leaks from the connectors. Make sure that the threads and internal body of all fittings are clean.

Before cylinders are transported, reduce internal pressure to 5–25 psig. It is recommended to place a nitrogen pad of less than 25 psig on the cylinder prior to

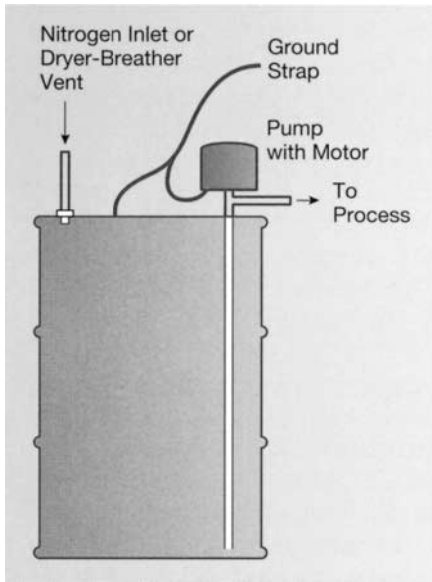


FIG. C-36 Drum unloading system. (Source: ARCO Chemical.)

the return shipment. The shipping regulations permit freight-forwarding and common carriers to charge a rate higher than normal if pressure is above 25 psig, since that places the tank in a "Compressed Gas" category.

Unloading drums

Follow all applicable safety procedures. Be sure full protective clothing is worn (see Fig. C-40) when opening the drum plug (bung), when placing or operating pumps, or when flushing out empty drums. In the event of spillage, see "Handling Spills and Leaks" below.

If the TDI is frozen, or if there is a possibility of freezing because the drums have been exposed to ambient temperatures below 17°C (63°F), then the drums should be heated to 35–43°C (95–110°F) until all TDI is liquid. Do not heat above 43°C (110°F). After the TDI is thawed, the drums should be rolled for at least 30 min to uniformly mix the 2,4- and 2,6-isomers.

During unloading, drums should be kept under a nitrogen pad to prevent contamination by water vapor. However, unloading by pressure is unsafe.

The preferred method is by pump, either manual or electric (see Fig. C-36). If the pump is electrical, be sure the drum is properly grounded. If the drum is to be unloaded by gravity, the faucets should be self-closing. Bungholes should be fitted with a dryer-breather vent device to prevent drum collapse.

Thawing TDI

Thawing TDI in tank cars

TDI is shipped in insulated tank cars. During the winter, it is loaded at temperatures between 38 and 43°C (100–110°F). Despite these precautions, there may be substantial heat loss before the car reaches its final destination. Therefore, during the winter, all incoming tank cars of TDI should be checked for freezing. The 2,4-isomer of TDI-80 freezes at 15°C (59°F), the 2,6-isomer at 7.2°C (45°F). Between these two temperatures, only the 2,4-isomer freezes. If this happens, isomer stratification takes place.

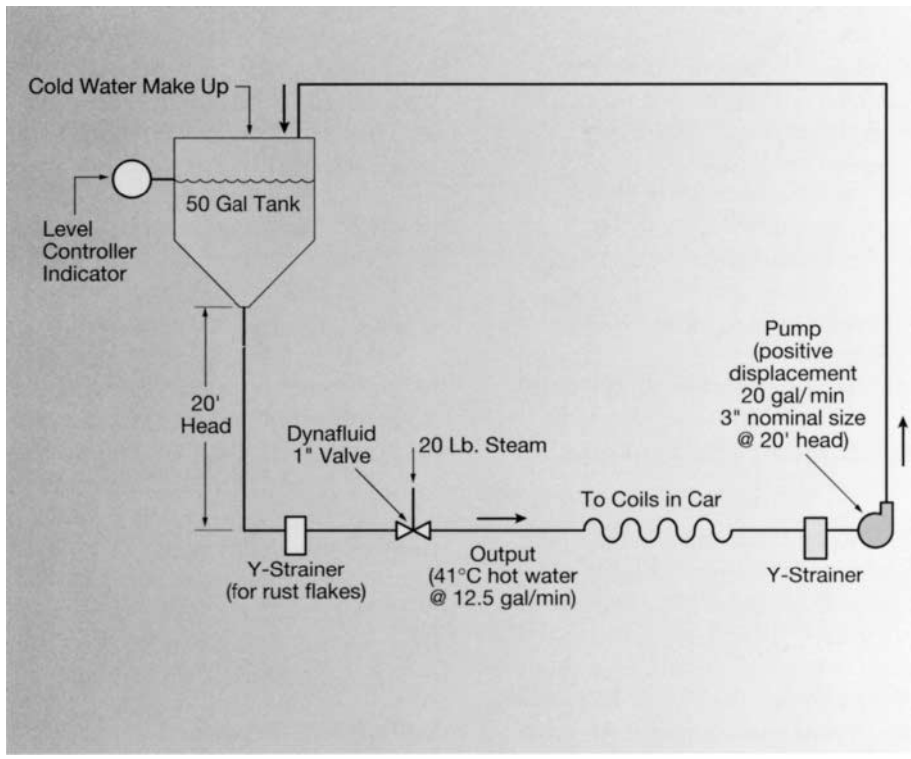


FIG. C-37 Steam/water mixing system. (Source: ARCO Chemical.)

Note: After thawing TDI, the layers remain separated. If they are not mixed, processing problems can be expected. However, if proper care is taken in thawing and remixing TDI, the quality can be maintained and no processing problems should occur.

How to determine if TDI is frozen. The way to tell if TDI is frozen is by taking its temperature while wearing proper protective equipment. Do not open the manway to inspect it visually. Temperature measurement is accurate and will detect frozen TDI, even when it is not visible.

When to heat a TDI tank car. If the TDI temperature is less than 17°C (63°F), the car should be heated before it is unloaded.

Note: If the car is not to be heated immediately, it should be repressurized to 5–10 psig with nitrogen to prevent crystals from forming as the result of contamination of the TDI with water. It should be depressurized before heating and unloading.

How to heat a TDI tank car. The TDI should be heated to 35–43°C (95–110°F) until all the frozen TDI has thawed. Never allow the TDI temperature to exceed 43°C (110°F). If TDI is overheated, dimerization may take place. (See discussion under *Heat* above and graph showing conditions for dimer formation, Fig. C-27.) If dimer forms, the TDI should not be used.

Heat Sources: The best way to thaw frozen TDI is with tempered hot water, thermostatically controlled to 41°C (106°F). Hot water is less likely to cause dimerization than steam. If tempered hot water is not available, an alternate source of heat is 20-lb steam, mixed with cold water. A steam/water mixing system similar to the one shown in Fig. C-37 can be used to obtain the desired temperature.



FIG. C-38 Steam hose connections. (Source: ARCO Chemical.)

Plants that have only steam available should avoid pressures above 20 lb. High-pressure steam, if not watched very carefully, will rapidly overheat the TDI. Even at lower temperatures, careful monitoring must take place.

Heat Source Connections: Tank cars were designed by different tank car manufacturers and put into service at different times. Therefore, cars must be carefully examined to determine the size and location of the external coil inlets and outlets.

In general, the inlet is on one side of the car, away from the handbrake (Fig. C-38). Some cars have two inlet valves. On these cars, the one farthest away from the handbrake side is for the left-side coils; the one nearest the handbrake side is for the right-side coils.

After TDI is thawed. After the TDI has been heated to 35–43°C (95–110°F), it must be completely mixed to eliminate isomer separation. Unload the entire contents into a bulk storage tank and circulate for 2–3 h before use.

Thawing TDI in cylinders

TDI will freeze at temperatures below 60°F. It is therefore imperative that during winter, cylinders be stored in a temperature-controlled environment. Recommended storage temperature is 70°F.

However, if the product does freeze, each cylinder must be placed in a heated room. The material should be completely thawed prior to use.

During this time period, daily movement of the cylinder will be necessary to allow the TDI isomers to thoroughly mix inside the cylinder. Short, jerking motions while moving with a forklift will provide sufficient agitation. To avoid product damage, never apply steam or an open-air flame to the exterior of the cylinder. A nitrogen pad of 20–25 psig should be maintained while the cylinder is being stored or heated.

Storage of TDI

TDI may be stored indoors or outdoors.

If TDI is stored indoors, the building should have sprinklers, good ventilation, and adequate heat to maintain storage temperature of 21°C (70°F). Constant monitoring of TDI temperature is required. If TDI is stored outdoors, or if indoor



FIG. C-39 Typical TDI storage tank. (Source: ARCO Chemical.)

temperature may drop below 21°C, provisions must be made for warming and thawing the TDI. These include adequate tank and line insulation, external heating coils or jackets, and steam-traced or electrically heated lines.

If thawing is necessary, never heat the TDI above 43°C (110°F). Prolonged overheating will cause dimer formation (see *Heat* above). After thawing, mix the TDI to eliminate isomer separation. Use a tank agitator or a circulating pump.

Whether indoors or outdoors, bulk storage tanks should be blanketed with nitrogen. Without this dry atmosphere, water vapor will react with the TDI to form solid aromatic polyurea, which can plug lines and foam machine heads.

A pneumatic bubbler gauge[1] that operates with nitrogen is recommended. This gauge measures the pressure required to displace TDI from a vertical tube in the tank.

Storage tank design

Vertical, cylindrical steel tanks (Fig. C-39) are normally preferred for storing TDI, although limited indoor headroom may dictate the use of horizontal tanks.

Storage tanks may be field-erected on a concrete foundation, and there is no practical limitation to size. Recommended capacity is 30,000 gal for tank car deliveries and 6–8000 gal for tank trucks. In other words, capacity should be sufficient to accept the entire contents of a tank car or truck, even when half-filled. The storage tank vent should be routed to an approved emission control system.

Materials of construction

TDI tanks can be made from carbon steel (ASTM A 285 Grade C) or from stainless steel (Type 304 or 316). API Code 650 specifies $^1/_4$-in steel for the bottom and $^3/_{16}$-in for the shell and roof.

Stainless steel tanks require no lining and are recommended. Carbon steel may also be used provided it is rust-free, sandblasted, and "pickled" with an initial TDI charge prior to use, or has a baked phenolic lining. Recommended are: Heresite P 403,[2] Lithcote LC 73,[3] Amercote 75,[4] or Plascite 3,070.[5] The inside surface should

[1] Petrometer Corp. or Varec Div., Emerson Electric Co.

[2] Heresite-Saekaphen, Inc.

[3] Lithcote Company.

[4] Amercon Corporation.

[5] Wisconsin Protective Coatings.

be sandblasted to a commercial finish and cleaned prior to the application of the lining.

Hose and piping to receive TDI

From Tank Cars: TDI is discharged by nitrogen pressure supplied by the customer through flexible hose into piping to the storage tank. Both the hose and the piping are provided by the customer. The hose should be a polypropylene-lined flexible hose.

When unloading, it is also necessary to repressurize the car. Use a $^3/_4$-in reinforced rubber hose attached to the 1-in inert gas inlet fitting.

From Tank Trucks: TDI is usually discharged from a built-in compressor or pump on the truck, through flexible polypropylene-lined hose provided by the trucker, into piping supplied by the customer. The length of the hose is specified by the customer with the first order. The piping should be Schedule 316 stainless steel. An oil-and-water separator and pressure regulator are also suggested as an assembly in the pressure line off the compressor.

Auxiliary equipment

Valves: Ball valves should be stainless steel with nonvirgin TFE seals. Plug valves and gate valves are not acceptable. Valves may be threaded or they may be flanged (150-lb ASA or MSS).

Liquid Filter and Pressure Gauges: A filter should be placed in the piping between the tank car or tank truck and the storage tank. A cartridge with a 20- or 30-micron glass fiber element is recommended.

Pressure gauges should be installed on either side of the filter to measure the drop. This will indicate when the filter must be cleaned or replaced.

Sampling Valves: If delivery is by tank car, an in-line sampling valve is recommended.

Pumps: Sealless magnetic drive pumps are recommended for TDI transfer.

TDI Safety and Handling

The following contains information as of December 1997. The health and safety information is partial. For complete, up-to-date information, obtain and read the current *Material Safety Data Sheet* (MSDS). (To order an MSDS, call the chemical company's nearest office.)

TDI is a toxic and highly reactive compound. It should be kept in closed, isolated systems and transferred with care. However, TDI is not a difficult material to handle. If proper procedures are followed, there is relatively little chance of danger.

The sections below briefly discuss some possible hazards and describe what to do in an emergency. Plant personnel should be thoroughly familiar with these procedures.

Reactivity hazards

TDI is a stable compound with a relatively high flash point. However, it will react with water, acids, bases, and other organic and inorganic compounds. TDI is also affected by heat and, like any organic compound, will burn.

Water: When TDI comes in contact with water, aromatic polyurea is formed, heat is generated, and carbon dioxide is evolved. Pressure buildup from the carbon dioxide will occur. This pressure could rupture a storage vessel. To help prevent reactions with water, the TDI should be kept under a nitrogen pad.

Chemical: Contact between TDI and acids should be avoided. Contact with bases, such as caustic soda and primary and secondary amines, might produce a violent

reaction. The heat given off causes pressure buildup and risk of rupture of the storage vessel. Contact with tertiary amines (commonly used as urethane catalysts) may cause uncontrollable polymerization, with a similar result. High temperatures may also cause dimerization.

TDI should be kept away from certain rubber and plastics. These materials will rapidly become embrittled; cracks may develop and their strength may be weakened.

Fire hazards

TDI has a flash point of 132°C (270°F) and therefore does not constitute a severe fire hazard. However, TDI is an organic material and will burn when exposed to fire. In addition, the flash point of TDI does not reflect the hazards presented by any cellular or foam plastic product that contains TDI.

Health hazards

TDI is highly toxic through inhalation and if inhaled in significant quantities can produce serious health effects. TDI is an animal carcinogen and is considered to be a possible human carcinogen. TDI has a characteristically pungent odor. However, TDI is considered to have poor warning properties; if you can smell it, the concentration of TDI would be in excess of the occupational exposure limit of 0.005 ppm (0.04 mg/m^3) as an 8-h time-weighted average.

Inhalation: Repeated overexposure and/or a high one-time accidental exposure to TDI may cause allergic lung sensitization similar to asthma. Symptoms may include wheezing, choking, tightness in the chest, and shortness of breath. Any individual exposed to TDI above the occupational exposure limit may develop these symptoms; however, for sensitized persons, these symptoms may occur at or below the occupational exposure limit. Repeated overexposure to TDI may also produce a cumulative decrease in lung function.

Dermal and Oral Exposure: The liquid and vapor of TDI can cause moderate to severe irritation to the eyes, skin, and mucous membranes. If not rinsed off immediately (within 5 min), burns to the eyes and skin may occur with the possibility of producing visual impairment. While the oral toxicity of TDI is low, ingestion of TDI can result in severe irritation to the gastrointestinal tract and produce nausea and vomiting.

Protective clothing

Because of the health hazards associated with TDI, full protective clothing and equipment (see Fig. C-40) must be worn whenever there is a possibility of contact. Such occasions include, but are not limited to:

- Opening tank car hatches, truck manway covers or drum plugs
- Connecting and disconnecting hoses and pipes
- Placing and operating pumps
- Breaking TDI piping, including piping previously decontaminated
- Flushing (cleaning) TDI drums
- Pouring foams, in operations where ventilation may not be adequate

Where liquid TDI spills can occur, butyl rubber clothing should be worn. If any article of clothing becomes contaminated, it should be removed immediately and discarded promptly.



FIG. C-40 Protective clothing and equipment. (Source: ARCO Chemical.)

The odor warning of TDI is insufficient to be used as a method for detecting the presence of hazardous concentrations. Whenever there is a chance that airborne levels of TDI vapors could exceed the recommended Threshold Limit Value (0.005 ppm as an 8-h time-weighted average or 0.02 ppm as a ceiling value), a NIOSH/MSHA positive-pressure, supplied-air respirator should be worn. When issuing respirators to employees, follow all OSHA respirator requirements (29 *Code of Federal Regulations* 1910.134).

The equipment necessary to properly protect any individual who may come into contact with liquid TDI is shown in Fig. C-40.

Emergency Actions

The following section contains basic information on what to do in the event of an accident.

In addition, the Chemical Manufacturers Association (CMA) has established CHEMTREC to give advice on spill, leak or fire emergencies involving transportation or transport equipment. The current CHEMTREC number for the United States and Canada is 800-424-9300.

In the District of Columbia or from outside the U.S., call 703-527-3887.

Note: If the spill is greater than 100 lb, U.S. federal law requires it to be reported to the National Response Center (NRC). The number is 800-424-8802.

First aid

If there is known contact with TID, take the following steps:

Eye Contact: Flush the eyes with clean, lukewarm water; then periodically flush for 20–30 min. Prompt medical attention should be sought.

Skin Contact: Immediately flush thoroughly with water for 15 min. Seek medical attention if ill effect or irritation develops.

Inhalation: Immediately move victim to fresh air. Symptoms of exposure to TDI vapors include: tightness in the chest, watering eyes, dry throat, nausea, dizziness, and headaches. The onset of symptoms may be delayed, so a doctor should monitor exposed personnel.

Handling spills and leaks

Wear a NIOSH/MSHA-approved, positive-pressure, supplied-air respirator. Follow OSHA regulations for respirator use (see 29 *Code of Federal Regulations* 1910.134). Wear recommended personal protective equipment: clothing, gloves, and boots made of butyl rubber.

Spill and leak cleanup:

1. Stop the source of spill. Stop the spread of spill by surrounding it with dry noncombustible absorbent.
2. Apply additional dry noncombustible absorbent to the spill. Add approximately 10 parts decontamination solution to every one part spilled TDI.

Suggested Formulation for Decontamination Solution

	% by Weight
Water	75
Nonionic Surfactant[a]	20
n-propanol	5

[a] e.g., Poly-Tergent® SL-62 (Olin).

3. Sweep up material and place in proper DOT-approved container. Use more decontamination solution to clean remaining surfaces and also place this residue in container.
4. Loosely apply lid. Do not seal for 48 h, since gas generation may occur during neutralization. Isolate container in a well-ventilated place.
5. Discard all contaminated clothing. Decontaminate personnel and equipment using approved procedures.

Decontamination of empty containers:

1. Spray or pour 1–5 gal of decontamination solution into the container. Ensure that the walls are triple rinsed.
2. Leave container standing unsealed for a minimum of 48 h to allow for a complete neutralization of TDI.

Disposal:

1. Care should be taken to prevent environmental contamination from the use of this material.
2. Dispose of contaminated product, empty containers and materials used in cleaning up leaks, spills, or containers in a manner approved for this material.
3. The user of this material has the responsibility to dispose of unused materials, residues, and containers in compliance with all relevant federal, state, and local laws and regulations regarding treatment, storage, and disposal for hazardous and nonhazardous wastes.
4. Ensure that drums are labeled with correct hazardous waste code. Waste code U223.

Chillers; Crystallizers; Chemical Separation Method; Alternative to Distillation/Fractional Distillation*

Crystallization: An Alternative to Distillation

Many organic mixtures may be separated by cooling crystallization. In simple terms, cooling crystallization means that a mixture of organic chemicals is partially crystallized by reduction in temperature, without removal of any of the components by evaporation. An example of a crystallizer is illustrated in Fig. C-41.

Crystallization is a one-way process: the heat is removed, crystals are formed, and the mixture of crystals and solids are then separated. Many crystallizations take place at near ambient temperature so there is little heatup or cooldown required to get the right conditions for the separation to start.

Distillation, on the other hand, is a refluxing operation, where products are repeatedly evaporated and recondensed. Most distillations take place at elevated temperatures, which means that the materials being processed must be heated up and cooled back down again, usually with energy losses both ways. Also many distillations are run under vacuum to achieve better separation, which is energy intensive.

The latent heat of fusion in crystallization is generally much lower than the latent heat of vaporization. Since the latent heat must be removed only once, instead of many times as in distillation, the energy requirements are drastically lower for crystallization.

In the great majority of crystallizations, the crystals that form are 100 percent pure material, as opposed to something only slightly richer than the feed material

* Source: Armstrong Engineering Associates, USA. Adapted with permission.

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